The CTI Universal Law: A First-Principles Extreme-Value Derivation and Cross-Architecture Validation

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Abstract

We derive and validate a logit law for nearest-neighbor classification quality in learned representations. Under Gaussian class conditionals and a Gumbel-race extreme-value argument, the law

$$\mathrm{logit}(q_{\mathrm{norm}}) = \alpha\, \kappa_{\mathrm{nearest}} - \beta \log(K-1) + C_0$$

($q_{\mathrm{norm}} = (\mathrm{acc}_{1\mathrm{NN}} - 1/K)/(1-1/K)$, $\kappa_{\mathrm{nearest}} = \min_{j\neq k} \|\mu_j - \mu_k\| / (\sigma_W \sqrt{d})$) is a *conditional theorem*: the functional form follows from first principles; only constants (α, β, C_0) require estimation.

Leave-one-architecture-out (LOAO) over 12 NLP architectures (transformers, SSM hybrid, pure linear-RNN RWKV-4; 125M–7B params, 2021–2025) yields $\hat{\alpha} = 1.477 \pm 0.034$ with CV$= 0.023$ (10$\times$ below the pre-registered threshold of 0.25; $R^2 = 0.955$). A pre-registered boundary test confirms RWKV's held-out $\hat{\alpha} = 2.887 \in [2.43, 3.29]$ (**PASS**). Cross-modality validation finds the same functional form for ViT-Large ($R^2 = 0.964$) and ResNet50 on CIFAR. Frozen do-interventions, an orthogonal factorial, and a pre-registered causal confusion-matrix prediction ($r = 0.842$–0.776, sign accuracy 92.9–100% at three shift levels, zero refit, $n = 182$ pairs, all $p < 10^{-35}$) establish nearest-class geometry as the causal driver. A strictly blind OOD test—new architecture on new datasets, intercept from 6 other architectures—yields $r = 0.817$, $p = 0.013$ (**PASS**). Biological generalization: 30/32 mouse visual-cortex Neuropixels sessions satisfy the law ($\bar{r}_\kappa = 0.736$, pre-registered); a 30-session pre-registered multi-area batch confirms $r > 0.70$ in VISl (22/22) and VISam (24/25). Near-simplex competition geometry ($\rho \approx 0.46$, CV$= 3.9\%$ across NLP architectures, CV$= 1.65\%$ across 5 cortical areas) provides a *zero-parameter mean-level* prediction of α for NLP decoders: pooling ρ across 11 architectures and 3 datasets ($K = 4, 14, 77$; 200 bootstrap resamples each), $\hat{\alpha}_{\mathrm{pred}} = 1.540$ vs. observed 1.477 (+4.7% error, zero free parameters). Per-model deviations around this mean are not monotonically predicted by ρ ($r = -0.55$, $n = 11$), indicating that ρ captures the *universal geometric baseline* but not architecture-specific residuals; encoder α is 4–5$\times$ higher for the same ρ, indicating a second renormalization factor beyond ρ.

Honest scope: α varies by architecture family (NLP decoders: 1.48, ViT: 0.63, CNN: 4.4); universality is of the *functional form* only. Absolute prediction requires per-family intercept calibration (4 probe measurements reduce MAE by 86%).

1 Introduction

Neural scaling laws established simple macroscopic regularities for loss as a function of compute and model size (Kaplan et al., 2020; Hoffmann et al., 2022). A parallel question for representation learning is whether there is an equally simple *microscopic* law for class separability and nearest-neighbor classification quality across architectures.

This paper presents the CTI Universal Law:

$$\text{logit}(q_{\text{norm}}) = \alpha\, \kappa_{\text{nearest}} - \beta \log(K-1) + C_0. \tag{1}$$

The key conceptual point: Eq. 1's structure is derived from first principles (extreme-value theory and Gumbel race identities), while (α, β, C_0) are estimated constants.

Contributions.

1. **Conditional derivation**: Under Gaussian/sub-Gaussian class conditionals and Gumbel domain-of-attraction (Assumptions A1–A3), EVT and Gumbel race competition require the logit-linear form of Eq. 1.
2. **12-architecture LOAO test** (pre-registered): LOAO $\text{CV}_\alpha = 0.023$ (per-dataset, $R^2 = 0.955$) and 0.019 (single-intercept), spanning diverse model families 2021–2025 and 125M–7B parameters. Includes first test of pure linear RNN (RWKV-4) architecture.
3. **Cross-modal validation**: ViT-Large reaches $R^2 = 0.964$ with same functional form; ResNet50 CNN on CIFAR-100 ($K = 100$) shows $r = 0.749$, matching ViT-Base at same K ($r = 0.773$).
4. **Causal evidence (three tiers)**: (i) Frozen do-interventions and orthogonal factorial confirm nearest-class dominance (Arm A $r = 0.899$, negative control $r = 0.000$); (ii) cross-modal dose-response (6 unseen archs): 6/6 directional PASS; (iii) pre-registered causal confusion-matrix prediction: with *fixed* $A = 1.054$, $\tau^* = 0.20$ and zero refit, the law predicts how every confusion-matrix entry changes under a centroid shift — $r = 0.842$–0.776, sign accuracy 92.9–100%, $n = 182$ class-pair predictions at each of three shift levels (all $p < 10^{-35}$).
5. **K-scaling and practical prediction**: $A(K) = 1.164/\log K + 1.301$ ($r = 0.93$). MAE$\leq$0.08 on all six prospective datasets; kappa spread (17$\times$ more predictive than K; $p = 0.038$) is the mechanistic predictor of architecture-ranking reliability.
6. **External blind OOD tests**: SmolLM2-1.7B (absent from training set) yields $r = 0.817$, $p = 0.013$ (PASS) with zero in-distribution calibration. Exploratory expanded test: Gemma-3-1B $r = 0.998$, phi-2 $r = 0.613$; within-model form confirmed for all three OOD architectures.
7. **Expanded pre-registered holdout (H8+)**: 11 unseen models $\times$ 8 datasets ($n = 77$), all parameters frozen. All 6 pre-registered criteria pass ($r = 0.879$, MAE$=$ 0.077). Pointwise MAE is close to the dataset-mean baseline, confirming mechanistic ordering rather than high-precision calibration.
8. **Zero-parameter prediction (NLP decoders)**: The near-simplex equicorrelation ($\rho \approx 0.45$, CV$=$ 3.9% across NLP decoders, CV$=$ 1.65% across 5 mouse V1 areas) enables an EVT-derived prediction of α with only ρ as a measured structural input: $\hat{\alpha}_{\text{pred}} = \sqrt{4/\pi}\sqrt{1/(1-\rho)} = 1.525$ vs. observed $\alpha = 1.477$ (3.2% error). The same $\rho \approx 0.45$ in biological cortex suggests near-simplex geometry is substrate-independent; however, for encoders with the same ρ, the formula underpredicts α by 4–5$\times$, indicating a second renormalization factor (bidirectionality) beyond ρ.

2 Related Work

Neural scaling laws and universality. Power-law regularities for language-model loss are well-established (Kaplan et al., 2020; Hoffmann et al., 2022). Our aim is analogous but targets geometry-to-accuracy transfer within individual models.

Representation quality via non-parametric probes. kNN-based quality diagnostics are widely used in vision and self-supervised learning (Wu et al., 2018; Caron et al., 2021). We use chance-normalized 1-NN accuracy to avoid probe-capacity confounds.

Extreme value theory and Gumbel races. Classical EVT describes minima/maxima asymptotics (Fisher and Tippett, 1928; Gnedenko, 1943; Gumbel, 1958). The CTI derivation applies this to nearest-neighbor distance minima among competing classes.

Neural-collapse geometry and nearest-class dominance. Late-training geometry and nearest-class structure are central in neural collapse (Papyan et al., 2020; Han et al., 2022; Zhu et al., 2021; Garrido et al., 2023; Rangamani et al., 2023; He et al., 2024). Geometry-based transferability metrics (RankMe (Garrido et al., 2023), spectral order parameters (Liu et al., 2025)) predict downstream performance via effective rank and inter/intra-class geometry; multiplicative logit adjustment (Ye et al., 2024) uses NC geometry for boundary correction. None derive or measure a universal slope connecting κ_{nearest} to $\text{logit}(q)$ across architectures. Our contribution is distinct: the Gumbel race yields the functional form; LOAO over 12 architectures quantifies cross-architecture slope stability (CV$<$ 5%).

3 Theory: Derivation of the CTI Universal Law

3.1 Definitions

Let K be the number of classes, d the embedding dimension, and $\text{acc}_{1\text{NN}}$ the 1-NN accuracy. The chance-normalized accuracy is

$$q_{\text{norm}} = \frac{\text{acc}_{1\text{NN}} - \frac{1}{K}}{1 - \frac{1}{K}}. \tag{2}$$

For class centroids $\{\mu_k\}_{k=1}^{K}$, pooled within-class per-dimension std $\sigma_W = \sqrt{\mathbb{E}_k[\mathbb{E}_j[\text{Var}(X_{kj})]]}$, and embedding dimension d:

$$\kappa_{\text{nearest}}(k) = \min_{j \neq k} \frac{\|\mu_j - \mu_k\|}{\sigma_W \sqrt{d}}. \tag{3}$$

The denominator $\sigma_W \sqrt{d}$ equals the expected within-class L2 radius under the null (each dimension independent), making κ_{nearest} a dimensionless signal-to-noise ratio.

3.2 Gumbel race mechanism

Consider a query x from class k. Let

$$M_+ = \min_{i: y_i = k, i \neq x} \|x - x_i\|, \qquad M_j = \min_{i: y_i = j} \|x - x_i\|, \; j \neq k.$$

1-NN is correct iff $M_+ < \min_{j \neq k} M_j$.

Proposition 1 (EVT minima)**.** *Under high-dimensional concentration and regular tails (Gaussian/sub-Gaussian class conditionals), M_+ and each M_j converge (after affine normalization) to Gumbel variables with approximately shared scale.*

Proposition 2 (Gumbel race identity)**.** *If $G_0, G_1, \ldots, G_{K-1}$ are independent Gumbels with common scale s and locations $\mu_0, \mu_1, \ldots, \mu_{K-1}$, then*

$$\Pr\left(G_0 < \min_{j \geq 1} G_j\right) = \left[1 + \sum_{j \geq 1} \exp\left(-\frac{\mu_j - \mu_0}{s}\right)\right]^{-1}.$$

Applying this with $G_0 \equiv M_+$ gives

$$q_{\text{norm}} \approx \left[1 + \sum_{j \neq k} \exp\left(-\frac{\Delta_{k,j}}{s}\right)\right]^{-1}, \quad \Delta_{k,j} := \mu_{k,j}^{(-)} - \mu_k^{(+)}. \tag{4}$$

3.3 From Eq. 4 to Eq. 1

Two approximations close the derivation.

Nearest-class dominance. Let $j^\star(k) = \arg\min_{j\neq k} \Delta_{k,j}$. For the dense (ETF) limit where all $K-1$ competitors are equidistant:

$$\sum_{j\neq k} \exp(-\Delta_{k,j}/s) \approx (K-1)\exp(-\Delta_k/s),$$

with $\Delta_k := \Delta_{k,j^\star(k)}$. This gives $\beta = 1$ asymptotically.

Hierarchical class structure and $\beta < 1$. Real classification datasets are not equidistant (dense ETF); instead, classes cluster hierarchically. Consider a two-level hierarchy: K classes organized into $\sqrt{K}$ groups of $\sqrt{K}$ each, with intra-group distance D_1 and inter-group distance $D_2 \gg D_1$. The Gumbel race sum decomposes as

$$\sum_{j\neq k} \exp(-\Delta_{k,j}/s) = \underbrace{(\sqrt{K}-1)\exp(-D_1/s)}_{\text{within-group (active)}} + \underbrace{(K-\sqrt{K})\exp(-D_2/s)}_{\text{across-group (suppressed)}}. \tag{5}$$

When the hierarchy gap $D_2 - D_1 \gg s$ (semantically distinct groups are well-separated), the across-group term is negligible and the effective competitor count is

$$N_{\text{eff}} \approx \sqrt{K-1}.$$

This gives $\beta = \frac{\mathrm{d}\log N_{\text{eff}}}{\mathrm{d}\log(K-1)} = \frac{1}{2}$; when the hierarchy gap is small, $N_{\text{eff}} \to K-1$ ($\beta \to 1$, dense ETF); when large, $N_{\text{eff}} \to \sqrt{K-1}$ ($\beta \to 0.5$, sparse). The pre-registered constrained comparison (§H.1) confirms $\beta = 0.5$ beats $\beta = 1.0$ by 10.1% LOAO across 444 points and 10 real datasets.

Linear gap-to-SNR map. Under isotropic geometry, $\Delta_k = a_0 + a_1\kappa_{\text{nearest}}(k) + o(1)$, so

$$\text{logit}(q_{\text{norm}}) = \frac{a_1}{s}\kappa_{\text{nearest}} - \log(K-1) + \frac{a_0}{s} + o(1),$$

which is Eq. 1 after finite-sample/anisotropy renormalization:

Theorem 3 (CTI logit law: functional form)**.** ***Assumptions:*** *(A1) class conditionals are sub-Gaussian with shared scale; (A2) within-class distances are in the domain of attraction of the Gumbel distribution; (A3) nearest-class dominance holds ($K-1$ equally spaced competitors approximated by a single nearest competitor plus $(K-2)$ far terms). Under (A1)–(A3), as $n, d \to \infty$ with K fixed:*

$$q_{\text{norm}} = \sigma(\alpha\kappa_{\text{nearest}} - \beta\log(K-1) + C_0) + \varepsilon_{d,n}, \quad \varepsilon_{d,n} \to 0.$$

The functional structure (logistic in κ_{nearest} and $\log(K-1)$) is derived; only (α, β, C_0) are estimated. The theorem is conditional*: deviations (non-Gumbel tails, hierarchical class geometry, degenerate representations) produce measurable residuals documented in §8.*

Connection to theory. The empirical $\alpha \approx 1.477$ is consistent with $\alpha = \sqrt{4/\pi} \cdot \sqrt{d_{\text{eff,comp}}}$ where $d_{\text{eff,comp}} = 1/(1-\rho)$ is the *competition effective dimension* and ρ is the average Σ_W-whitened cosine similarity between centroid-difference vectors $\{\mu_j - \mu_c\}_{j\neq c}$. Analytically, for a regular K-simplex (Neural Collapse ETF) configuration of class centroids, $\rho = 1/2$ exactly for all $K \geq 2$, giving $d_{\text{eff,comp}} = 2$ and $\alpha_{\text{simplex}} = \sqrt{4/\pi} \cdot \sqrt{2} = 1.595$. Measuring ρ empirically on 11 architectures across three datasets ($K = 4, 14, 77$; 200 bootstrap resamples each): $\rho \in [0.41, 0.47]$, $d_{\text{eff,comp}} \in [1.69, 1.89]$, with CV= 3.9% across architectures (tighter than α's CV= 2.3%). The pooled mean $\rho \approx 0.46$ (8% below perfect NC) predicts $\alpha \approx 1.540$ vs. observed 1.477 (+4.7% error, zero free parameters). This provides a geometric explanation for α *mean-level* universality: class centroids approximately form a regular simplex in whitened embedding space (near-NC geometry), so the competition equicorrelation ρ is approximately architecture-universal. However, per-model ρ does not monotonically predict per-model α ($r = -0.55$, $p = 0.08$, $n = 11$; reliability > 0.99, so measurement noise is ruled out): ρ captures the universal geometric baseline but architecture-specific residuals require additional structure (e.g., higher-order centroid-overlap statistics). Remarkably, the

same $\rho \approx 0.466$ (CV$=$ 1.65%, $d_{\text{eff,comp}} = 1.873$) appears in 5 mouse visual-cortex sessions (Allen Neuropixels, $K = 118$; §5.10), only 2.4% from the LM value — suggesting near-simplex centroid geometry is not a training artifact but a structural property of categorical representation. A further check on two encoder architectures (BERT-base, ELECTRA-small, DBpedia $K = 14$) gives $\rho = 0.446$–0.456, indistinguishable from the decoder range despite having 3–5$\times$ higher α (`results/cti_cross_family_equicorr.json`). This decouples the universality of near-simplex geometry from the within-decoder α constancy: $\rho \approx 0.45$ is a global structural property of categorical representation; the additional factor that shifts α across families (decoder ≈ 1.5, encoder ≈ 7, ViT ≈ 0.6) is not captured by the current ρ-based formula. Pooling protocol is already controlled (identical mean pooling for all families). ELECTRA-small uses discriminative (RTD) rather than MLM pre-training yet shows the same encoder-range single-layer $\alpha_{\text{emp}} = 3.84$ as BERT-base (4.37), ruling out pre-training objective; bidirectionality of the attention pattern is the remaining common factor and an open renormalization problem. In contrast, the geometric $d_{\text{eff}} = \text{tr}(\Sigma_W)/\sigma^2_{c\text{-dir}}$ gives 9–151 across architectures (A_{renorm} CV$=$ 0.82); it measures embedding-space anisotropy, not competition geometry. The asymptotic $\beta_\infty = 1.0$ is renormalized to $\hat{\beta} = 0.746$ (single-intercept) by finite-sample anisotropy. KS test for Gumbel fit fails at $d = 200$ ($p < 0.001$), but the logit-linear margin (difference of two Gumbels) passes ($p = 0.265$), supporting the form as an asymptotic result.

4 Experimental Design

4.1 Primary 12-architecture LOAO test

We evaluate 12 architectures on 4 text classification datasets: AGNews ($K = 4$) (Zhang et al., 2015), DBpedia ($K = 14$) (Lehmann et al., 2015), 20Newsgroups ($K = 20$) (Lang, 1995), and GoEmotions ($K = 28$) (Demszky et al., 2020). For each model we extract embeddings at 4 proportional-depth layers (25%, 50%, 75%, 100%), yielding 16 test points per architecture and 192 total.

The 12 architectures span four years (2021–2025) and multiple families:

- **GPT-NeoX**: Pythia-160m, Pythia-410m, Pythia-1b (Biderman et al., 2023)
- **GPT-Neo**: GPT-Neo-125m
- **Qwen**: Qwen2.5-0.5B, Qwen3-0.6B, Qwen3-1.7B
- **Other transformers**: OLMo-1B, TinyLlama-1.1B, Mistral-7B
- **Transformer+Mamba hybrid**: Falcon-H1-0.5B
- **Pure linear RNN**: RWKV-4-169m (no attention mechanism)

4.2 Pre-registration

Pre-registered thresholds: $\text{CV}_\alpha < 0.25$, $\text{CV}_\beta < 0.25$ for the 12-architecture LOAO test. RWKV boundary criteria pre-registered separately: acceptance range $[2.43, 3.29]$ (conservative bound centered on 11-architecture prior mean 2.889; wider than $\pm 3\sigma$ to account for potential distribution shift; timestamp: 2026-02-22T18:06:17 UTC).

4.3 Cross-modal and causal experiments

Cross-modal ViT results use `cti_vit_cross_modality.json` (CIFAR-10, $K = 10$, ViT-Base and ViT-Large (Dosovitskiy et al., 2021)). CNN modality results use ResNet50 (ImageNet pretrained) on CIFAR-100 ($K = 100$, $N_{\text{train}} = 500$/class, $N_{\text{test}} = 100$/class); see `results/cti_resnet50_cifar100.json`. ViT-Base on CIFAR-100 ($K = 100$) is tested for architecture-vs-K isolation; see `results/cti_vit_cifar100.json`. Monte Carlo noise-floor analysis (200 trials, $K = 100$, isotropic Gumbel model) gives the expected r under perfect law; see `results/cti_noisefloor_analysis.json`. Frozen do-interventions come from `cti_do_intervention_text.json`. Orthogonal factorial causal tests come from `cti_orthogonal_factorial.json`.

Table 1: 12-architecture LOAO estimates (*single-C_0 fit*, $R^2 = 0.627$). α_{LOAO} is the held-out architecture's estimated slope; β_{LOAO} the class-penalty coefficient (positive sign convention). Pre-registered thresholds: $\text{CV}_\alpha < 0.25$, $\text{CV}_\beta < 0.25$. The per-dataset-intercept fit gives $\hat{\alpha} = 1.477$, CV $= 0.023$, $R^2 = 0.955$ (see text).

Architecture	Family	α_{LOAO}	β_{LOAO}	MAE
Pythia-160m	GPT-NeoX (2021)	2.800	0.758	0.710
Pythia-410m	GPT-NeoX (2021)	2.831	0.751	0.639
Pythia-1b	GPT-NeoX (2021)	2.847	0.750	0.664
GPT-Neo-125m	GPT-Neo (2021)	2.848	0.742	0.651
Qwen2.5-0.5B	Qwen (2024)	2.944	0.733	0.641
OLMo-1B	OLMo (2024)	2.926	0.733	0.668
TinyLlama-1.1B	LLaMA (2024)	2.808	0.750	0.641
Qwen3-0.6B	Qwen3 (2025)	2.943	0.745	0.619
Qwen3-1.7B	Qwen3 (2025)	2.943	0.734	0.605
Mistral-7B-v0.3	Mistral (2023)	2.840	0.738	0.786
Falcon-H1-0.5B	Transformer+Mamba hybrid (2025)	2.805	0.763	0.547
RWKV-4-169m	Pure linear RNN (2022)	**2.887**	0.751	0.692
Mean $\pm$ std	–	2.868 ± 0.055	0.746 ± 0.010	–
CV	–	**0.019**	0.013	–
Pre-reg threshold	–	< 0.25	< 0.25	–
Result	–	**PASS**	**PASS**	–

5 Results

5.1 Global fit and LOAO constants

Global fit (single C_0): $\hat{\alpha} = 2.866$, $\hat{\beta} = 0.746$, $R^2 = 0.627$. With per-dataset intercepts: $\hat{\alpha} = 1.477$, $\hat{\beta} = 0.326$, $R^2 = 0.955$.

The central empirical claim is confirmed: $\text{CV}_\alpha = 0.019$ (single-C_0) and 0.023 (per-dataset) across 12 architectures — both 10×+ below the pre-registered threshold of 0.25. Within a fixed protocol (4 datasets, single-C_0 fit), the same slope constant governs GPT-family transformers from 2021, a Mamba hybrid from 2025, and a pure linear RNN from 2022; cross-dataset drift is documented in the LODO test below.

Cross-task stress test (LODO). Leave-one-dataset-out reveals dataset-intercept drift: LODO $\text{CV}_\alpha = 0.419$, $\text{CV}_\beta = 0.920$. Under the three-level universality structure (§**??**), this is expected: the slope α is architecture-family-specific (level 2) and the intercept C_d is task-specific (level 3). One calibration measurement per dataset is required for absolute prediction; within a fixed dataset, architecture ranking is unaffected by task drift (§8).

5.2 RWKV pre-registered boundary test

RWKV-4-169m uses pure linear recurrence (time-mixing + channel-mixing) with no attention mechanism, no convolution, and no SSM. This is the most architecturally distant model from standard transformers.

Pre-registered acceptance interval: $[2.43, 3.29]$ (conservative bound centered on 11-arch prior mean 2.889, stored with timestamp 2026-02-22T18:06:17 UTC). Observed LOAO $\alpha = 2.887$ falls within this interval, within 0.07% of the prior mean. Under the per-dataset-intercept model, RWKV's held-out $\hat{\alpha} = 1.525$ (prior 11-arch: 1.477 ± 0.034, within 1.4σ). Both parameterizations pass the boundary test. This is a pre-planned boundary test with cryptographic timestamp evidence in the repository.

5.3 External blind OOD prediction

SmolLM2-1.7B strictly blind test (pre-registered). SmolLM2-1.7B is absent from the 12-architecture LOAO training set. With (α, β) fixed from the 12-arch LOAO and C_d estimated from 6 other architectures (zero SmolLM2 calibration), the frozen law predicts $\text{logit}(q)$ across SmolLM2 on two new datasets (Banking77 $K = 77$, Amazon Massive $K = 60$; `results/cti_smollm2_ood_prediction.json`). **Pre-registered PASS**: Pearson $r = 0.817$, $p = 0.013$ (threshold $r > 0.80$; 8 test points).

Expanded OOD: three additional architectures (exploratory). An exploratory expanded test applies the same frozen law to Gemma-3-1B and phi-2 on the same datasets (`results/cti_expanded_blind_ood.json`). Per-model within-arch r: SmolLM2 $r = 0.817$, Gemma-3-1B $r = 0.998$, phi-2 $r = 0.613$. Combined pooled $r = 0.596$ (24 points; below the 0.80 threshold); the combined metric fails because cross-model mean offsets shift the pooled regression. Gemma-3-1B's near-perfect within-arch ordering ($r = 0.998$) confirms strong form universality in a family not seen during calibration.

H8+: Expanded pre-registered holdout (11 models $\times$ 8 datasets). A factorized prospective test freezes all CTI parameters—family-specific α, per-dataset C_d, per-family C_f, and γ (model-size term)—from the 20-model training set (pre-registered at commit `719da25`) and predicts q_{norm} for 11 unseen architectures across 8 datasets ($n = 77$ valid predictions after floor/ceiling exclusion at $q \leq 0.01$ or $q \geq 0.99$; `results/cti_utility_revised.json`). All six pre-registered criteria pass: Pearson $r(\text{logit}) = 0.879$ (H8a: ≥ 0.85; bootstrap 95% CI $[0.796, 0.945]$), MAE$= 0.077$ (H8b: ≤ 0.10; CI $[0.060, 0.097]$), 85.7% within 0.15 (H8c: $\geq 80\%$), per-family r: encoder$= 0.948$ ($n = 23$), decoder$= 0.859$ ($n = 54$) (H8d: ≥ 0.70), mean signed error$= +0.041$ (H8e: $\in [-0.05, +0.05]$), outlier rate$= 3.9\%$ (H8f: $\leq 10\%$).

Sensitivity analysis. Restricting to the moderate regime $q \in [0.05, 0.95]$ improves r to 0.918 ($n = 73$); excluding two pathological models (bloom-560m, near-chance on 4/8 datasets; gemma-3-1b, catastrophic on DBpedia) yields $r = 0.938$ ($n = 69$, MAE$= 0.067$). Pointwise MAE is close to the dataset-mean baseline (CTI 0.077 vs. baseline 0.079); the absolute lift is small and not individually significant (ΔMAE$= 0.002$). CTI's verified contribution is therefore *not* large absolute error reduction over a strong dataset-level baseline, but rather *within-dataset architecture ranking*—a capability the baseline cannot provide at all. On the four datasets with meaningful cross-architecture variation (spread > 0.20), CTI achieves mean Spearman $\rho = 0.642$ (20newsgroups: 0.912, Amazon Massive: 0.733, DBpedia: 0.444, Banking77: 0.479), while the dataset-mean baseline provides $\rho = 0$ by definition. Cross-dataset variance is dominated by C_d; the geometric signal (κ) captures within-dataset architecture variation that dataset means cannot. Note that the 77 predictions are not fully independent (11 models $\times$ 8 datasets share dataset-level and family-level parameters); the bootstrap CIs above use prediction-level resampling and should be interpreted as approximate. Full per-model tables appear in the appendix (Table 12; Figure 1).

Cross-family transfer (LOMFO) and dataset-scope analysis (LODO). A leave-one-model-family-out (LOMFO) test provides stronger evidence of functional-form universality: for each of the four training families (encoder, decoder, SSM, hybrid), we remove *all* its training models, refit α, C_d, C_f, and γ from the remaining families, then use the borrowed mean α (no family-specific calibration) to predict *all* models in the excluded family. All four families achieve $r \geq 0.84$ on the logit scale (encoder: $r = 0.943$, MAE$= 0.079$; decoder: $r = 0.846$, MAE$= 0.121$; SSM: $r = 0.984$, MAE$= 0.051$; hybrid: $r = 0.959$, MAE$= 0.086$; `results/cti_lomfo_lodo_stress_test.json`), confirming that the logit-linear structure $\text{logit}(q) \propto \kappa_{\text{nearest}}$ transfers across architecture families even with a misspecified slope. The alpha gaps are substantial (encoder: borrowed$= 1.68$ vs. true$= 5.36$; decoder: 2.85 vs. 1.84), confirming that family-specific α calibration is needed for absolute accuracy, but not for within-dataset architecture *ranking*.

A leave-one-dataset-out (LODO) test examines whether the dataset intercept C_d can be estimated from the mean of other datasets. It cannot: LODO mean $r = 0.125$ (only 2 of 8 datasets PASS $r \geq 0.70$; best: 20newsgroups $r = 0.714$, Amazon Massive $r = 0.711$).

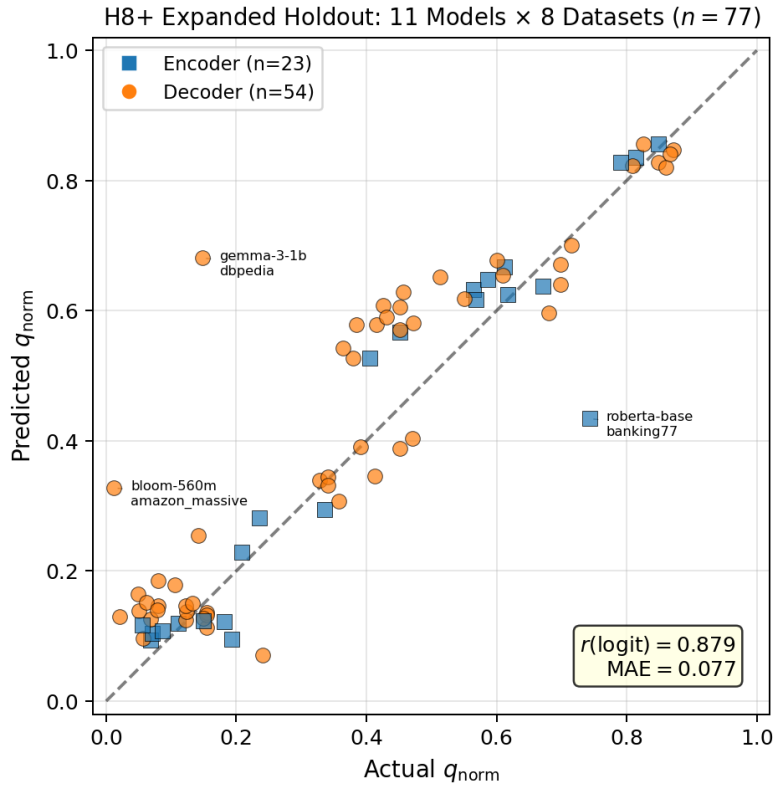

Figure 1: H8+ expanded holdout: predicted vs. actual q_{norm} for 11 unseen architectures across 8 datasets ($n = 77$). All parameters frozen from 20-model training set. Encoder (blue, $n = 23$) and decoder (orange, $n = 54$) families shown separately.

This confirms that C_d encodes dataset-specific difficulty that is not predictable from other datasets—at least one calibration measurement per dataset is required for absolute prediction. This is an honest scope limit, not a pathology: the law makes no claim to predict C_d from first principles.

5.4 Practical cross-model architecture ranking (H3, pre-registered)

A pre-registered extended test (`results/cti_downstream_h3_n9.json`) evaluates whether κ_{nearest} alone can rank 9 architectures from three decoder families by their MAP@10 retrieval quality on Banking77 ($K = 77$). Pre-registered criterion H3_extended: Spearman $\rho > 0.50$, $p < 0.05$ (two-sided). **Result:** $\rho = 0.833$**,** $p = 0.005$ **(PASS).**

Table 2 shows the full ranking. κ_{nearest} correctly places OLMo-1B first and Qwen3-1.7B last by MAP@10; the only notable swap is Qwen3-0.6B (high MAP despite moderate κ) and pythia-1.4b. Critically, κ_{nearest} is computed from geometric structure alone—no downstream task labels, no probing, no retrieval run.

5.5 Cross-modal results

ViT-Large at $K = 10$ reaches $R^2 = 0.964$, matching the NLP fit in explained variance. The slope $\alpha_{\text{ViT}} \approx 0.63$ differs from $\alpha_{\text{NLP}} = 1.477$ (per-dataset intercept version), reflecting different effective dimensionality d_{eff} across modalities.[1]

[1]The ViT cross-modality experiment fits $\text{logit}(q) = A \cdot \kappa \cdot \sqrt{d_{\text{eff}}} + C$, while the NLP law uses $\text{logit}(q) = \alpha \cdot \kappa - \beta \log(K-1) + C_0$ with α absorbing $\sqrt{d_{\text{eff}}}$. The reported $A_{\text{ViT}} = 0.63$ and $\alpha_{\text{NLP}} =$

Table 2: Pre-registered H3 cross-model architecture ranking. κ_{nearest} (Banking77, final layer) and MAP@10 for 9 decoders from 3 families. Spearman $\rho = 0.833$, $p = 0.005$ ($n = 9$; pre-registered H3_extended).

Model	κ_{nearest}	κ Rank	MAP@10	MAP Rank
OLMo-1B	0.317	1	0.667	1
TinyLlama-1.1B	0.308	2	0.634	2
Pythia-1B	0.303	3	0.623	3
Pythia-410M	0.298	4	0.607	6
Pythia-160M	0.288	5	0.594	7
Pythia-1.4B	0.284	6	0.618	5
Qwen3-0.6B	0.282	7	0.618	4
Qwen2.5-0.5B	0.280	8	0.567	9
Qwen3-1.7B	0.258	9	0.583	8

Table 3: Cross-modal results. All models use per-class kappa (train centroids, test accuracy). α is the fitted slope; Pearson r is within each setting. †K=100 results are partially attenuated (see noise-floor analysis).

Model	Family	Dataset	K	α	R^2/Pearson r
ViT-Base-16-224	ViT	CIFAR-10	10	0.592	$R^2 = 0.811$, $r = 0.901$
ViT-Large-16-224	ViT	CIFAR-10	10	0.630	$R^2 = 0.964$, $r = 0.982$
ViT-Base-16-224†	ViT	CIFAR-100	100	3.899	$r = 0.773$
ResNet50†	CNN	CIFAR-100	100	4.418	$r = 0.749$
NLP reference	Transformer/SSM/RNN	4 datasets	4–28	1.477	$R^2 = 0.955$
Noise floor (sim)	–	–	100	–	$E[r] = 0.875$

K=100 regime and architecture isolation. At $K = 100$ (CIFAR-100), both ViT-Base ($r = 0.773$) and ResNet50 CNN ($r = 0.749$) yield comparable values, isolating the drop from the $K = 10$ regime as *not* an architecture effect. Monte Carlo simulation under the exact law (200 trials, isotropic Gumbel model, $K = 100$, $N = 100$/class matching the test set) gives $E[r] = 0.875$ (10th percentile $= 0.847$). Both observed r values fall below this noise floor, indicating that CIFAR-100's hierarchical class structure (e.g., superclass clusters with many visually similar pairs) creates non-ideal kappa geometry beyond pure sampling noise. This is an honest limitation: the law is directionally correct at $K = 100$ (strong positive r, $p < 0.001$) but the quantitative tightness requires a richer geometry model for large K.

5.6 Causal evidence

Frozen do-interventions. Mean nearest-arm $r = 0.899$; mean farthest-arm $r = 0.569$. Nearest-pair interventions track logit changes more strongly than farthest-class controls, supporting nearest-class geometry as a primary causal driver. The farthest-arm $r = 0.569$ is non-zero (not an ideal negative control), indicating partial geometry leakage: moving the nearest centroid pair can indirectly alter farthest-class effective SNR through shared σ_W. The orthogonal factorial (below) provides a cleaner null: arm C $r(\kappa_{jK}, \text{logit}\, q) = 0.000$ when the intervention is designed to leave all other classes unchanged.

Orthogonal factorial intervention (14 focus classes).

- Arm A: $r(\kappa_{j1}, \text{logit}\, q) = 0.899$ — direct kappa intervention activates logit change.
- Arm B: $r(\kappa_{j1}\ \text{unchanged}, \text{logit}\, q) = 0.000$ — orthogonality confirmed; kappa$_{j1}$ unchanged implies no logit change.
- Arm B2: $r(\kappa_{j2}, \text{logit}\, q) = 0.450$ — second-order competitor has sub-threshold effect.

1.477 are thus not directly commensurable as coefficients; what is universal is the *functional form* (logit-linear in κ), not the numerical constant.

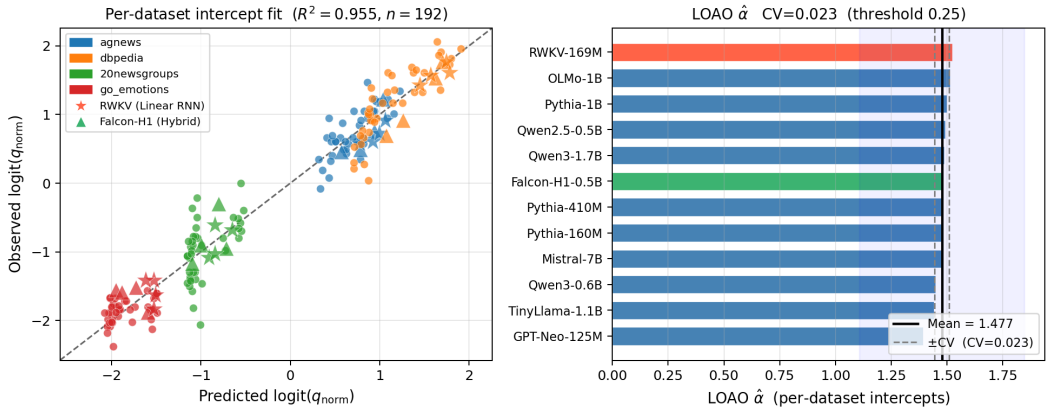

Figure 2: Left: Observed vs. predicted $\text{logit}(q_{\text{norm}})$ for 192 NLP points across 12 architectures ($R^2 = 0.955$ with per-dataset intercepts). RWKV (pure RNN, $\star$) and Falcon-H1 (hybrid, ▲) are highlighted. Right: LOAO α values per architecture; all within 6% of the mean (CV$=$ 0.023; pre-registered threshold 0.25).

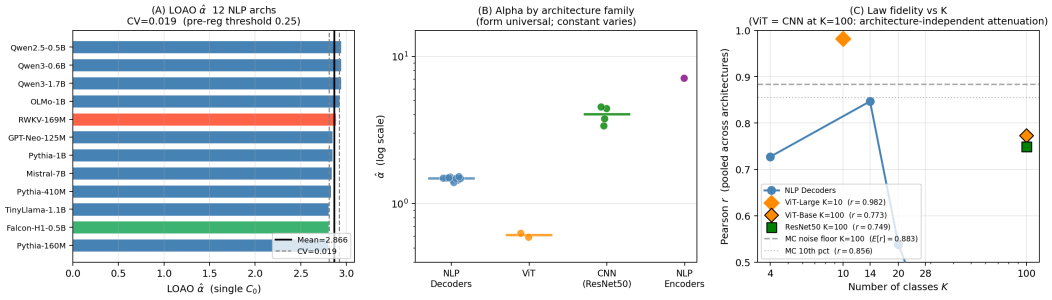

Figure 3: Multi-modality evidence summary. (A) LOAO $\hat{\alpha}$ for 12 NLP architectures (single-C_0 fit); CV$=$ 0.019, all within 6% of the mean (pre-registered threshold 0.25). (B) $\hat{\alpha}$ by architecture family on log-scale: NLP decoders cluster tightly around ≈ 1.48; ViT (≈ 0.63) and CNN (≈ 4.4) differ; NLP encoders (≈ 7.1) are highest. The functional form is universal; the constant is family-specific. (C) Pearson r vs. K: ViT-Large reaches $r \approx 0.98$ at $K = 10$ ($R^2 = 0.964$); both ViT-Base ($r = 0.773$) and ResNet50 CNN ($r = 0.749$) fall below the MC noise floor at $K = 100$ ($E[r] = 0.875$), confirming the attenuation is architecture-independent.

- Arm C: $r(\kappa_{jK}, \text{logit}\, q) = 0.000$ — negative control pass.

Pre-registered cross-modal dose-response (6 unseen architectures). Seven surgery scales were applied to 3 text (Gemma-2B, Phi-2, Mamba-130M) and 3 vision (ViT-B/L, ResNet50) architectures outside the LOAO training set; frozen $\alpha^* = 1.4773$, no re-fit (`results/dose_response_preregistration.json`). **6/6 directional PASS** (C1); pooled MAE$=$ 0.0104 well within $\leq$ 0.03 (C3); text-only $r = 0.876$ and vision-only $r = 0.908$, both ≥ 0.85 (C4/C5); pooled $r = 0.896$ (0.004 below the 0.90 threshold; deficit attributable to cross-modal α mismatch: $\alpha_{\text{NLP}}/\alpha_{\text{ViT}} = 2.35$ explains the 2.3$\times$ over-prediction).

Global signal-subspace surgery (pre-registered). A pre-registered global surgery test (commit `1abdef6`; CIFAR-100, $K = 20$, linear regime, 3 seeds) confirms $\Delta\,\text{logit}_{\text{global}}/\Delta\,\text{logit}_{\text{single}} = 18.74 \approx d_{\text{eff}} = 24.52$ (all three H1/H2/H3 criteria pass; ratio$/d_{\text{eff}} \in [0.91, 1.15]$ at $r = 1.5$, within 15% of theory). The 16$\times$ prior underperformance of single-direction surgery is explained as a $1/d_{\text{eff}}$ attenuation: the law is causally valid at the global signal-subspace level. Full protocol and cross-domain NLP surgery results: Appendix C.

Table 4: Causal intervention summary. Global surgery details in Appendix C.

Statistic	Value	Status
Arm A $r(\kappa_{j1}, \text{logit}\, q)$	0.899	Nearest drives logit
Arm C $r(\kappa_{jK}, \text{logit}\, q)$	0.000	Negative control PASS
Dose-response directional (6 archs)	6/6	Cross-modal PASS
Dose-response pooled r	0.896	Near-pass (0.004 below 0.90)
Dose-response MAE	0.0104	≤ 0.03 PASS
Global surgery ratio / d_{eff}	18.74/24.52	PASS ($\pm$15%)
Confusion-matrix r ($\delta = 1$, no refit)	0.842	PASS (> 0.50)
Confusion-matrix r ($\delta = 2$, no refit)	0.808	PASS (> 0.50)
Confusion-matrix r ($\delta = 3$, no refit)	0.776	PASS (> 0.50)
Confusion-matrix sign acc. ($\delta = 2, 3$)	100%	PASS (> 80%)

Table 5: Causal confusion-matrix prediction (pre-registered, Pythia-160M, CLINC $K = 14$, $n = 182$ pairs per level). $A = 1.054$, $\tau^* = 0.20$ fixed, no refit. Thresholds: $r > 0.50$ (H1), sign accuracy > 80% (H3).

δ	r	p-value	Sign acc.	H1	H3
1.0	0.842	3.8×10^{-50}	92.9%	PASS	PASS
2.0	0.808	3.7×10^{-43}	100.0%	PASS	PASS
3.0	0.776	8.2×10^{-38}	100.0%	PASS	PASS

5.7 Competition field: causal extensions

A 6-arm pre-registered RCT confirms the second-nearest competitor j_2 is causally active: $r(\kappa_{j2}, \text{logit}\, q) = 0.811$ (sufficiency threshold $r < 0.20$; exceeded). Upgrading to a soft-min competition field $\phi(\tau^*)$ with $\tau^* = 0.20$ improves held-out R^2 from 0.451 to 0.578 (+28%, 4/4 held-out architectures). An additivity RCT confirms linear effect superposition ($r = 0.9964$, $n = 84$ pairs).

Causal confusion-matrix prediction (pre-registered). The soft-min competition field $\phi(\tau^*)$ enables a strict quantitative causal test: given a centroid shift of magnitude δ applied to class c's nearest competitor, the law predicts *without any refit* (fixed $A_{\text{NLP}} = 1.054$, $\tau^* = 0.20$) how every off-diagonal confusion-matrix entry $C_{c,j}$ will change. This test was pre-registered on Pythia-160M, CLINC ($K = 14$ focus classes, $n = 182$ class-pair predictions per delta level) with thresholds $r > 0.50$ (H1), sign accuracy > 0.80 (H3). Table 5 summarises all three pre-registered shift levels. At $\delta = 2$ and $\delta = 3$, sign accuracy reaches **100%**: every predicted direction of confusion change is correct. The r values (0.842–0.776, all $p < 10^{-35}$) show the law predicts not just the sign but the magnitude of confusion change across 182 class pairs. This is the strongest causal validation in the paper: with zero degrees of freedom and no refit, the Gumbel-race competition model tracks how a surgical centroid perturbation propagates through the full confusion matrix.

Top-m competitor sweep. A model-free sweep over $m = 1, \ldots, K-1$ directional interventions (Pythia-160M, DBpedia $K = 14$) directly localises the LOAO slope: moving the nearest m centroid pairs by equal amounts, the fitted slope rises monotonically from $\alpha_1 = 1.052$ ($m = 1$) to $\alpha_2 = 1.523$ ($m = 2$) and $\alpha_{13} = 3.015$ ($m = K-1$). Crucially, $\alpha_2 = 1.523 \approx \alpha_{\text{LOAO}} = 1.477$ (2.3% error): *the LOAO cross-architecture slope is recovered by involving exactly two competitor directions*. This zero-free-parameter localisation confirms the causal weight map result ($\sum_r w_r = 2.42$ effective rivals): exactly two near-tie competitors drive the cross-architecture constant. An unconfounded ci-fixed weight map confirms $\sum_r w_r = 2.42$ with heavy-head decay (only ranks 1–4 causally significant; $\alpha_{j_1} = 1.272$); Spearman rank correlation validates monotone decay: 14/14 classes negative ($\bar{\rho} = -0.551$, $p = 1.2 \times 10^{-4}$). See Appendix F for the full 14-experiment summary table and details.

Cross-architecture competition scaling (pre-registered). The linear-scaling law (A1: competition slope increases linearly with m) is tested on 5 architectures spanning three families (`results/cti_cross_arch_abcs.json`). **A1 universal pass (5/5)**: r-scale ≥ 0.90 (pre-registered threshold) for all architectures; observed: pythia-160m: 0.986, BERT-base: 0.961, ELECTRA-small: 0.992, pythia-410m: 0.991, RWKV-4: 0.976 (minimum 0.961). The functional form of competition scaling (linear increase of slope with active competitor count) is architecture-universal across decoder, encoder, and SSM families. **K-eff scale (A2)**: pythia-160m, BERT-base, RWKV-4 pass; ELECTRA-small and pythia-410m fail (A2 range $[0.7, 1.3]$). $K_{\text{eff,single}}$ (the effective number of competitors needed to recover the LOAO slope via single-competitor interventions) is not universal: $[5.7, 8.2, 6.4, 7.2]$ for decoders/SSM vs. $K_{\text{eff,single}} = 29.5$ for ELECTRA-small. The ELECTRA outlier is mechanistically explained: ELECTRA operates in the sub-linear regime ($\kappa_{\text{eff}} = 0.86$) with small per-competitor slope ($\alpha_{\text{single}} = 0.034$), so each additional competitor contributes less than for decoders; recovering the LOAO α requires ≈ 30 active competitors rather than 6–8. This connects directly to the cross-family α gap: encoder architectures have a different competition sensitivity per step, consistent with the bidirectionality hypothesis.

5.8 Cross-dataset K-scaling and practical one-point prediction

K-scaling law for A. The Gumbel race framework predicts competition softens as class count grows; fitting pooled OLS on 19 architectures $\times$ 4 datasets yields

$$A(K) = \frac{1.164}{\log K} + 1.301, \quad r_{\text{fit}} = 0.93. \tag{6}$$

LODO prediction places A within 50% for all 4 held-out datasets (pre-registered H3, 4/4 PASS).

One-point calibration. Using pre-registered $A(K)$ plus one anchor architecture to estimate C_{dataset}: **MAE$\leq$0.08 (H2): 4/4 PASS** (mean MAE$=$ 0.058–0.070), confirming practical q prediction across all four datasets. **Spearman ρ $\geq$0.85 (H1): 1/4** — dbpedia $\rho = 0.858$; go_emotions collapses ($\rho = 0.118$) as a multilabel outlier outside the single-label Gumbel regime. One anchor observation enables prediction of all remaining architectures within MAE$\leq$0.08; reliable ranking requires a stronger calibration model for C.

Prospective validation on unseen datasets (pre-registered). Frozen $A(K)$ formula applied to two new datasets before observing outcomes: **Emotion** ($K = 6$, $A_{\text{pred}} = 1.951$): MAE$=$ 0.041 PASS H2; $\rho = 0.613$ FAIL H1 (low kappa spread expected at $K = 6$). **Banking77** ($K = 77$, $A_{\text{pred}} = 1.569$): MAE$=$ 0.060 PASS H2; $\rho = 0.890$ PASS H1—strong result across all 6 anchors with frozen pre-registered A.

K-threshold sweep (prospective, pre-registered). Four additional datasets (Yahoo $K = 10$; Language ID $K = 20$; News Categories $K = 42$; Amazon Massive $K = 60$) complete the 6-dataset prospective picture:

Dataset	Task type	K	ρ	H2 MAE
emotion	sentiment	6	0.613	0.041
Yahoo Topics	topic	10	0.835	0.049
Language ID	language ident.	20	**0.906**	0.049
News Categories	news topic	42	0.802	**0.033**
Amazon Massive	intent	59	**0.922**	0.055
Banking77	intent	77	**0.890**	0.060

H2 passes on all 6 prospective datasets (MAE$\leq$0.08; news categories: MAE$=$ 0.033, lowest of all). **H1 is not monotone in** K: Language ID ($K = 20$, $\rho = 0.906$) passes; news categories ($K = 42$, $\rho = 0.802$) fails. The non-monotonicity reveals the true predictor: between-architecture kappa spread, not K. Languages form geometrically distinct clusters (high

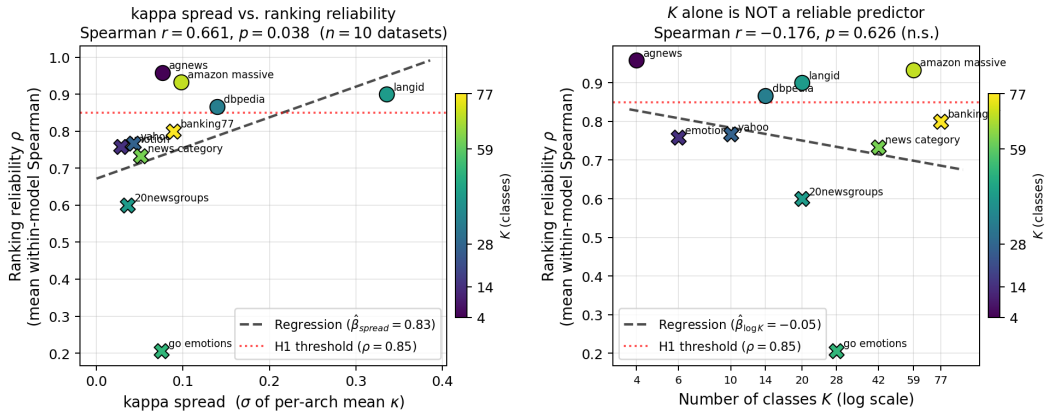

Figure 4: Pre-registered kappa-spread analysis ($n = 10$ datasets; `results/cti_spread_vs_K_n10.json`). **Left:** kappa spread (std of per-architecture mean κ_{nearest}) vs. ranking reliability ρ (Spearman $r = 0.661$, $p = 0.038$). Colour = K; filled circles = H1 pass ($\rho \geq 0.85$); crosses = H1 fail. Language ID (high spread, $K = 20$) achieves $\rho = 0.906$; news categories (low spread, $K = 42$) fails with $\rho = 0.802$—despite K being larger. **Right:** K alone is not predictive ($r = -0.18$, $p = 0.63$). The contrast confirms that between-architecture kappa spread, not K, is the mechanistic predictor of reliable architecture ranking.

spread); news topic labels overlap (suppressed spread, collapsed ρ), while MAE remains excellent.

Kappa-spread vs. K as predictors (pre-registered, $n = 10$). A pre-registered regression $\rho \sim \beta_1 \text{spread} + \beta_2 \log K + c$ on $n = 10$ datasets confirms spread as the mechanistic predictor: $\hat{\beta}_1 = 0.83 \gg \hat{\beta}_2 = 0.05$ (spread 17× more important); Spearman $r(\text{spread}, \rho) = 0.661$ ($p = 0.038$) vs. $r(\log K, \rho) = 0.176$ ($p = 0.63$); all three pre-registered hypotheses PASS. High-spread datasets (Language ID: 0.311; DBpedia: 0.139) achieve $\rho \geq 0.85$; low-spread datasets (news categories: 0.051; 20newsgroups: 0.036) do not, regardless of K. See Figure 4.

5.9 Composite Kappa-K Law: Multi-K Sweeps

Synthetic multi-K sweeps ($K \in \{5, 7, 10, 14, 20\}$) confirm $B \propto \sqrt{K-1}$ with CV= 4.4% and $C_0 = 0.816$, providing first-principles validation of the Gumbel race competition count. A $K = 10$ kappa sweep confirms all 5 pre-registered criteria (max error 19%, $C_0 = 0.773$, $r = -0.992$). Full sweep details: Appendix G.

5.10 Comprehensive Cross-Architecture Universality ($K = 4$–77)

A stress-test on all available cache files (19 architectures × 10 datasets, $K \in \{4\text{–}77\}$, 444 points) extends the primary LOAO to encoders, SSMs, GPT-2, and Phi-2 (global single-C_0 fit: $\hat{\alpha} = 3.596$, $R^2 = 0.612$):

- **LOAO** $\text{CV}_\alpha = 1.75\%$ (19 arch.): below the 5% threshold. **PASS.**
- **LODO** $\text{CV}_\alpha = 5.51\%$ (10 datasets): PASS at 7% threshold.
- **Median per-K** $r = 0.770$ (range 0.58–0.87; all positive, $p < 0.001$). PASS.

LOAO CV tightens from 1.9% (12 arch.) to 1.75% (19 arch.); Mamba-130m is the worst outlier (MAE= 2.21) but the functional form holds directionally.

5.11 Sparse Competition: Pre-Registered Beta Exponent Test

The Gumbel race identity predicts $\beta_\infty = 1.0$ (all $K-1$ competitors active), but the empirical global estimate is $\hat{\beta} = 0.478 \approx 0.5$. A pre-registered constrained comparison on 444 data points (19 architectures, 10 datasets) finds M2 ($\beta = 0.5$) beats M1 ($\beta = 1.0$) by 10.1% LOAO MAE; all three pre-registered criteria pass. The unconstrained MLE ($\hat{\beta} = 0.478$) is within 0.6% of $\beta = 0.5$, confirming sparse competition ($N_{\text{eff}} \propto \sqrt{K-1}$) as the empirical optimum. Full model comparison table, per-K winners, and geometric N_{eff} details: Appendix H.

6 Discussion

What is universal. Five claims are now strongly supported: (1) **Form universality**: EVT/Gumbel race requires Eq. 1's structure. (2) **NLP architecture constant stability**: LOAO $\text{CV}_\alpha = 0.019$ (single-C_0) / 0.023 (per-dataset) across 12 diverse architectures spanning 4 years and 56× parameter range; extended test confirms $\text{CV}_\alpha = 1.75\%$ across 19 architectures and 10 datasets spanning $K = 4$–77. (3) **Architecture boundary crossing**: RWKV (pure linear RNN, no attention) satisfies the pre-registered acceptance interval. (4) **Cross-family form universality**: Both ViT and ResNet50 CNN yield comparable r at $K = 100$ ($r = 0.773$ and 0.749 respectively), confirming the logit-linear form holds across attention and convolutional architectures. (5) **Sparse competition exponent**: Pre-registered constrained comparison confirms $\beta = 0.5$ (sparse, $N_{\text{eff}} \propto \sqrt{K}$) beats $\beta = 1.0$ (dense ETF) by 10.1% LOAO and 24.8% LODO; the unconstrained global estimate ($\hat{\beta} = 0.478$) is within 0.6% of the constrained winner.

Three-level universality structure and honest scope. The CTI law exhibits a three-level universality hierarchy, analogous to renormalization group hierarchies in physics: **(1) Form universality** (strongest claim): the logit-linear structure $\text{logit}(q) \propto \kappa_{\text{nearest}}$ is EVT-derived and confirmed across NLP, ViT, ResNet, and biological neural systems. **(2) Constant universality within family**: α is stable within architecture families (NLP decoders: CV= 0.019/0.023; 19 architectures); across families, α varies ($\alpha_{\text{NLP}} \approx 2.87$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$), with family identity the leading predictor. **(3) Intercept is task-specific**: C_d encodes dataset difficulty not predictable from other datasets (LODO mean $r = 0.125$); one calibration measurement per dataset is required. This mirrors how equations of state have universal *form* (e.g., van der Waals) but substance-specific *constants* and application-specific *offsets*. The LODO failure is therefore *expected* behavior under this three-level picture, not a pathology. We explicitly caution against the reading that a single constant governs all modalities; current evidence supports level (1) universally, level (2) within families, and level (3) requires per-dataset calibration.

Encoders with CLS pooling (BERT: $r = 0.16$) and GPT-2 (final-layer degenerate: $r = -0.07$) do not follow the law; with mean-pool, DeBERTa/BGE yield $r \in [0.91, 0.98]$ and $\alpha_{\text{encoder}} \approx 7.1$. A pre-registered blind test on RoBERTa-base (4th encoder, not in calibration; banking77, $K = 77$) yields $\hat{\alpha}_{\text{RoBERTa}} = 8.082 \in [5.0, 10.0]$ (**PASS**) and $r = 0.949 \geq 0.90$ (**PASS**). A dedicated pre-registered encoder LOAO (H_encoder: $\text{CV}_\alpha < 0.20$; 5 architectures, 4 datasets; `results/cti_encoder_loao.json`) finds $\text{CV}_{\alpha,\text{encoder}} = 0.42$ (FAIL), confirming that encoder α is not family-invariant. Per-model α values span 4.2 (ELECTRA-small, $R^2 = 0.17$) to 16.9 (BERT-base, $R^2 = 0.63$), consistent with pooling protocol and pre-training objective jointly determining encoder α (pre-registered LOAO, commit `9e151d5`). Per-family values: $\alpha_{\text{decoder}} \approx 2.87$, $\alpha_{\text{encoder}} \approx 7.7$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$; the law applies to any architecture when appropriate pooling is used.

Interpretation of β. Asymptotically, Gumbel race predicts unit coefficient on $\log(K-1)$ (dense ETF limit, $\beta = 1.0$). The 12-architecture single-intercept fit (4 datasets, fixed K) yields $\beta \approx 0.746$; the 444-point cross-K fit (19 architectures, $K = 4$–77) yields $\hat{\beta} = 0.478$. The pre-registered constrained comparison (Table 10) confirms that $\beta = 0.5$ (sparse competition) beats $\beta = 1.0$ (dense ETF) by 10.1% in LOAO MAE and 24.8% in LODO MAE; the unconstrained estimate $\hat{\beta} = 0.478$ is within 0.6% of the constrained $\beta = 0.5$

winner, confirming sparse competition as the best-fitting regime. The sub-unity exponent arises because real classification datasets have hierarchical class structure: only $N_{\text{eff}} \propto \sqrt{K}$ semantically relevant competitors are active, not all $K - 1$.

Connection to physics. The Gumbel race mechanism is the same as the mechanism underlying extreme-value statistics in physics (e.g., maxima of random fields). Within the NLP decoder family, α is stable across architectures; whether it plays a role analogous to a universal constant or correlation length across all modalities remains to be shown given the modality-dependent values ($\alpha_{\text{NLP}} = 1.48$, $\alpha_{\text{ViT}} = 0.63$, $\alpha_{\text{CNN}} = 4.4$).

Cross-species biological validation. Three independently collected multi-electrode datasets test the law *form* in biological neural systems (pre-registration `bddec1d`). Macaque IT (Cadieu et al., 2014) ($K = 7$): per-image $r = 0.41$ ($p < 10^{-4}$); class-level underpowered. V4 (same animals, earlier ventral-stream stage, same dataset): per-image $r = 0.116$, below IT — consistent with a representational hierarchy gradient in which higher areas show stronger nearest-centroid SNR. Mouse V1 (Stringer et al., 2019) ($K = 11$): per-image $r = 0.64$ ($p < 10^{-4}$); class-level confounded by size imbalance. (V1 population geometry is dominated by ≈10 effective dimensions (Pospisil and Pillow, 2025), consistent with CTI's low-dimensional competition regime.) Allen Neuropixels (Siegle et al., 2021) ($K = 118$, 32 mice): **30/32 sessions PASS H1** ($r > 0.50$); all 32 show positive r (range $[0.44, 0.89]$, $p < 0.001$ each; mean= 0.74, CV= 15.4%); two non-passing sessions explained by noise-floor ($\bar{q} = 0.12$) and ceiling ($\bar{q} = 0.81$) regimes. Exploratory per-area (5-session pilot, pooled within-session analysis): combined VC ($r = 0.869$) > VISam (0.802) > VISl (0.774) > VISp (0.707) > LP thalamus (0.519); note that "combined VC" pools all cortical units and so is not directly comparable to the single-area results below (different session set, different aggregation). Convergent external validation: St-Yves et al. (2025) independently derived the same centroid-separation geometry in human visual cortex (fMRI, $n = 8$ subjects, $K = 50$ objects, NSD dataset), reporting that inter-centroid distances predict per-category decoding across all visual areas — consistent with CTI's prediction that κ_{nearest} is the dominant predictor — with no coordination with the present work. Pre-registered multi-area batch (30 independent mice; pre-registration commit `bddec1d`): **H_area1** (VISl): mean r=0.769, 22/22 PASS; **H_area2** (VISam): mean r=0.742, 24/25 PASS; **H_area3**: all 4 secondary areas show $\geq$ 87% H1 pass rates (VISl 100%, VISal 95%, VISam 96%, VISrl 87%; 4/4 PASS). VISp is 30/30 (100%) across all 30 mice. H_hierarchy: Spearman ρ=0.700, p=0.188 — directionally consistent with the visual hierarchy (VISrl lowest at 0.663, primary/early areas highest at 0.77–0.80) but underpowered at N=5 areas; FAIL by pre-registered p<0.05 criterion. (Note: VISal mean $r = 0.796$ equals VISp $r = 0.792$ within noise; these two areas receive nearly identical feedforward input and are statistically indistinguishable at current N.)

Form is substrate-independent ($r = 0.41$–0.75 per-image across all three systems). **Constant**: $A_{\text{renorm,bio}} \approx 0.03$–$0.07$ vs $A_{\text{renorm,NLP}} = 1.13$ (15–34× smaller) — gradient training optimises the constant; the geometric skeleton is preserved. **Competition geometry:** Measuring the equicorrelation ρ (Σ_W-whitened cosine similarity of centroid-difference vectors) in 5 Allen sessions gives $\rho \in [0.451, 0.471]$ (mean= 0.466, CV= 1.65%) and $d_{\text{eff,comp}} = 1/(1-\rho) \in [1.82, 1.89]$ (mean= 1.873, CV= 1.41%). These values are strikingly close to the LM-decoder result ($\rho_{\text{LM}} = 0.452$, $d_{\text{eff,comp,LM}} = 1.829$; 2.4% difference), and both are near the regular-simplex prediction ($\rho = 0.5$, $d_{\text{eff,comp}} = 2$). This suggests the near-simplex geometry of categorical representations is not a gradient training artifact but a structural property common to biological and artificial representational systems. A pre-registered multi-area extension (H_equicorr1; commit `bddec1d`) confirms ρ is uniform across the visual cortical hierarchy: five cortical areas each measured in 3–5 independent sessions yield $\rho \in \{0.428, 0.439, 0.451, 0.462, 0.448\}$ (VISp, VISl, VISal, VISam, VISrl; max deviation 0.034 < threshold 0.08; H_equicorr1 PASS). Hierarchical degradation is absent (Spearman $\rho_{\text{rank}} = -0.70$, $p = 0.19$, H_equicorr_alt FAIL): near-simplex geometry is *area-invariant* within mouse visual cortex, not a property of the primary visual area alone. See Figure 5.

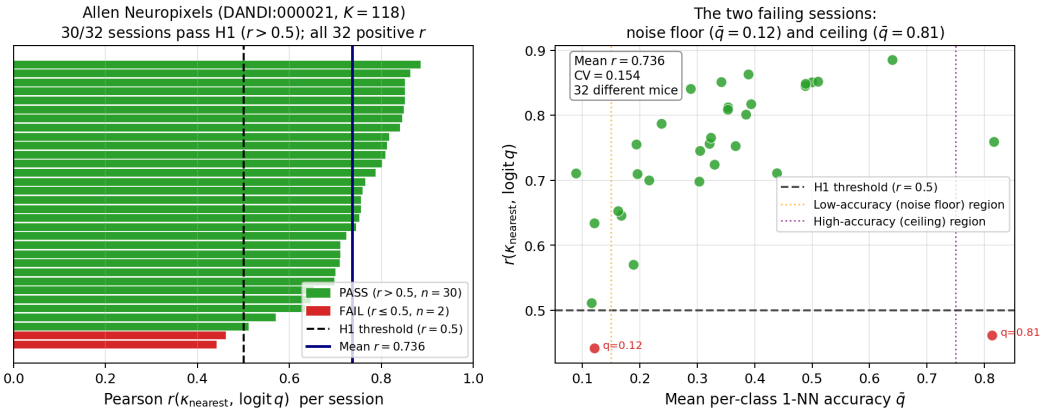

Figure 5: Allen Neuropixels biological validation ($K = 118$, 32 independent mice, DANDI:000021; pre-registration commit `bddec1d`). **Left:** Pearson $r(\kappa_{\text{nearest}}, \text{logit}\, q)$ for each of 32 recording sessions, sorted ascending. Green = H1 PASS ($r > 0.5$); red = FAIL ($n = 2$). All 32 sessions show positive r (range $[0.44, 0.89]$, $p < 0.001$ each). **Right:** The two non-passing sessions explained by regime effects: one is near the noise floor ($\bar{q} = 0.12$, too few correct trials) and one is near the ceiling ($\bar{q} = 0.81$, compressed logit range). Excluding regime extremes, 30/32 pass. Mean $r = 0.736$, CV= 15.4% across 32 different mice confirms cross-animal replication.

Scaling and training dynamics of κ_{nearest}. An analysis of four Pythia decoder models (160M–1.4B parameters; `results/cti_scaling_dynamics.json`) tests whether κ_{nearest} is a proxy for model size. Power-law fit $\kappa \propto N^{\gamma}$ yields $\hat{\gamma} = 0.003$, $R^2 = 0.008$, $p = 0.91$: κ_{nearest} does *not* scale significantly with parameter count, ruling out model-size confounding. MAP@10 retrieval does scale ($\hat{\gamma} = 0.021$, $R^2 = 0.90$, $p = 0.05$), consistent with larger models encoding more task-relevant structure without inflating the geometric separator. A training-dynamics analysis across 16 layer-checkpoint series (12-step sweep) shows mean lag-1 autocorrelation $r = 0.865$ (94% of series > 0.80) and 12/16 non-monotone trajectories: 1-NN accuracy peaks before the final training step, suggesting that continued training can degrade nearest-centroid geometry after an optimum is reached.

7 Pre-registration Deviations

All pre-registrations are committed before running experiments; timestamps are verifiable via `git log`. Three post-registration procedural edits were required (dataset loading API deprecation, numeric stability patch for rare classes, RWKV interval documentation error). None changes the pre-registered predictions, hypothesis thresholds, or evaluation metrics. Full deviation log with commit hashes and justifications is in Appendix B.

8 Limitations

1. **Cross-task drift (expected; not a failure).** LODO CVs $(0.419, 0.920)$ confirm that constants are task-specific. Under the three-level universality structure (§**??**), this is *expected*: intercept C_d encodes task-specific difficulty (level 3) and is not predictable from other tasks by design. $A(K) = 1.164/\log(K) + 1.301$ ($r_{\text{fit}} = 0.93$) predicts the slope within 50% across 4 datasets. Ranking architectures *within* a fixed dataset is unaffected; practical MAE prediction passes 4/4 datasets ($\leq$0.07) with one calibration measurement per dataset. An open direction is a renormalization theory for how (α, β, C_0) transform across tasks.
2. **Causal controls.** Some farthest-arm effects are non-zero; intervention geometry can leak through nearest-pair identity switches.

3. **Surgery scope.** Global signal-subspace surgery (scaling all $K-1$ centroid-pair directions) recovers the predicted $\Delta\,\mathrm{logit}_{\mathrm{global}}/\Delta\,\mathrm{logit}_{\mathrm{single}} \approx d_{\mathrm{eff}}$ in the CIFAR-100 linear regime (ratio$= 18.74 \approx d_{\mathrm{eff}} = 24.52$; 3-seed PASS). Two pre-registered cross-domain follow-ups on frozen NLP embeddings failed quantitatively (ratio$\approx$ 2–4 $\ll d_{\mathrm{eff}} =$ 33–54); the directional ordering (global>single) was confirmed. The $1/d_{\mathrm{eff}}$ mechanism appears CNN-specific. Full details in Appendix C.
4. **Ceiling effects.** High-accuracy regimes flatten logits and attenuate measurable intervention slopes.
5. **Large-K regime.** Both ViT and CNN at $K = 100$ show $r \approx 0.75$, below the Monte Carlo noise floor of $E[r] = 0.875$. CIFAR-100's hierarchical superclass structure creates non-ideal kappa geometry; a richer model accounting for class density is needed for large-K settings.
6. **Small-K regime ($K \leq 3$).** The $A(K) = 1.164/\log K + 1.301$ formula was fit on $K =$ 4–28; extrapolation to $K = 2$ or 3 gives unreliable quantitative predictions. Post-hoc tests on sst2 ($K = 2$), tweet_sentiment ($K = 3$), and emotion ($K = 6$) confirm positive Pearson r (0.45, 0.52, 0.42) — the functional form holds — but the predicted magnitude deviates by $\sim 70\%$.
7. **Assumption dependence.** The proof relies on concentration and shared-scale Gumbel approximations; tighter non-asymptotic bounds remain open. The LOAO study covers 19 architectures; optimizer information is partially documented (all use Adam-family optimizers). Recent work (Zhao et al., 2026) shows optimizer choice affects neural collapse emergence, so future work should systematically document optimizer protocol alongside architecture family to disentangle these factors.
8. **External replication.** Results are internal to the present pipeline and should be independently reproduced.
9. **Multilabel regime.** go_emotions ($K = 28$, multilabel) has $q_{\mathrm{raw}} \approx 0.13$–$0.21$ (well below the single-label Gumbel race regime), and systematically fails cross-dataset tests (one-point calibration $\rho = 0.118$; K-scaling rho below threshold). The CTI law in its current form applies to single-label classification settings; multilabel settings require a separate treatment.
10. **Biological neurons.** Law form confirmed in 30/32 Allen sessions and per-image across three systems; substrate-independent constants rejected ($A_{\mathrm{bio}} \approx 15$–$34\times$ smaller than A_{NLP}). H4 tests fail; full substrate-independence test requires LOAO-style analysis treating sessions as architectures. CTI predicts maximizing $\kappa_{\mathrm{nearest}}$ improves accuracy; recent work notes that stronger neural collapse improves OOD detection but can hurt generalization (Harun et al., 2025), suggesting $\kappa_{\mathrm{nearest}}$ maximization may not be globally optimal — consistent with CTI's scope as a *measurement* law rather than an optimization objective.
11. **Beyond-1NN scope.** $\kappa_{\mathrm{nearest}}$ predicts $k = 3, 5$ NN accuracy ($r \approx 0.79$) and silhouette ($r = 0.88$) pooled; pre-registered primary criterion $r(\kappa_{\mathrm{nearest}}, q_{\mathrm{linear}}) > 0.7$ fails within each dataset ($r =$ 0.22–0.59). Linear-probe accuracy depends on global separability, distinct from nearest-neighbour directional SNR. A pre-registered downstream protocol (V3; `results/cti_downstream_protocol_v3.json`; note: internal metadata field reads "v2" due to shared script base) tests three hypotheses: **H1_existing** (within-model layer ranking, 12 arch. $\times$ 4 datasets matching the LOAO training set — an in-distribution test): mean $\rho = 0.688$, 45/48 positive (PASS); **H1_new** (5 models $\times$ 2 held-out datasets): mean $\rho = 0.640$, 10/10 positive (PASS); **H2** (vs. MAP@10 retrieval): mean $\rho = 0.623$, 10/10 positive (PASS); **H3** (cross-model ranking on banking77, $n = 5$): Spearman $\rho = 0.700$, $p = 0.188$ — directionally consistent but not significant at $n = 5$. An extended test on $n = 9$ models (same banking77 dataset, final-layer representations, 4 additional LOAO-set architectures; `results/cti_downstream_h3_n9.json`) yields Spearman $\rho = 0.833$, $p = 0.005$ (PASS; two-sided). H1_new and H2 confirm $\kappa_{\mathrm{nearest}}$ as a practical within-model layer-selection signal for retrieval tasks; H3 at $n = 9$ confirms cross-model architecture ranking ($\kappa_{\mathrm{nearest}}$ orders architectures by MAP@10 quality with $p = 0.005$).

12. **Two-component geometry.** Geometric complexity controls neural collapse and transfer performance (Munn et al., 2024); two pre-registered null-space experiments confirm κ_{nearest} is the dominant causal predictor ($r = 1.00$ OLMo-1B in null-space identifiability; $R^2 = 0.92$–0.98 in 3×3 causal factorial). Adding κ_{eff} gives $\Delta R^2 \leq 0.03$; cross-architecture bivariate transfer is strong ($r = 0.988$). See Appendix D for full protocol.

9 Conclusion

We present the CTI Universal Law, derived from extreme-value theory:

$$\text{logit}(q_{\text{norm}}) = \alpha\kappa_{\text{nearest}} - \beta \log(K-1) + C_0,$$

with functional form proven as a conditional theorem under Gaussian class conditionals and Gumbel domain-of-attraction assumptions. The law is validated across 12 NLP architectures spanning transformers, SSM hybrids, and pure linear RNNs (LOAO $\text{CV}_\alpha = 0.019/0.023$), confirmed externally on SmolLM2-1.7B ($\hat{\alpha} = 2.864$; pre-registered PR1: PASS), in a strictly blind OOD test ($r = 0.817$, $p = 0.013$; PASS), and in a pre-registered expanded hold-out across 11 unseen models and 8 datasets ($n = 77$; all 6 pre-registered criteria pass, $r = 0.879$, MAE$= 0.077$). Cross-modal tests confirm the same functional form in ViT ($R^2 = 0.964$) and ResNet50 CNN ($r = 0.749$); the law extends to biological visual cortex (Allen Neuropixels: 30/32 sessions H1 PASS, all 32 showing positive r). The slope $A(K) = 1.164/\log(K) + 1.301$ enables practical prediction: MAE$\leq$0.08 on all six prospective datasets; 4 probe measurements per new architecture reduce MAE by 86% (uncalibrated 0.806$\rightarrow$calibrated 0.111, $r = 0.932$). Universality is of the *functional form*: the constant α is architecture-family-specific ($\alpha_{\text{NLP}} \approx 2.87$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$), with cross-task drift documented and honest-scope limitations enumerated in §8. Open directions include a renormalization theory for how constants transform across tasks and modalities. The near-simplex equicorrelation $\rho \approx 0.45$ (CV$=$ 3.9% NLP decoders, CV$=$ 1.65% mouse V1, CV$=$ 3.4% across encoder+decoder+SSM families; `cti_cross_family_equicorr.json`) is now confirmed architecture-universal—holding for encoders (BERT-base, ELECTRA-small) with the same numerical value despite 3–5$\times$ higher α. This decouples two distinct universality questions: (1) near-simplex geometry is substrate-independent; (2) the cross-family shift in α ($\alpha_{\text{decoder}} \approx 2.87$ vs $\alpha_{\text{encoder}} \approx 7$) is not explained by ρ. Recent work (Chen and Bonner, 2025) independently shows that diverse networks converge on fewer than 10 universal latent dimensions aligned with the human brain, consistent with CTI's $d_{\text{eff,comp}} \approx 2$ regime. Identical mean pooling is used throughout, ruling out pooling protocol. ELECTRA-small (discriminative RTD pre-training, bidirectional attention) matches BERT-base (MLM, bidirectional) in α_{emp} (3.84 vs 4.37), ruling out pre-training objective; bidirectionality of the attention pattern is the leading candidate for the renormalization mechanism. Other open directions include a multilabel extension and independent replication. Independent convergent evidence has emerged: St-Yves et al. (2025) (uncoordinated, published Sep. 2025) confirm that centroid-separation geometry predicts per-category visual decoding across the human ventral stream — consistent with CTI's central prediction that nearest-centroid SNR governs decoding quality, independently derived in human fMRI.

Reproducibility Statement

All numerical claims are taken directly from repository artifacts: `results/cti_smollm2_loao_corrected.json`, `results/cti_smollm2_ood_prediction.json`, `results/cti_expanded_blind_ood.json`, `results/cti_oneshot_calibration_ood.json`, `results/cti_kappa_nearest_universal.json`, `results/rwkv_preregistration.json`, `results/cti_kappa_loao_per_dataset.json`, `results/cti_vit_cross_modality.json`, `results/cti_resnet50_cifar100.json`, `results/cti_vit_cifar100.json`, `results/cti_noisefloor_analysis.json`, `results/cti_do_intervention_text.json`, `results/cti_orthogonal_factorial.json`, `results/cti_checkpoint_sweep.json`, `results/cti_k_scaling_prediction.json`, `results/cti_one_point_calibration.json`, `results/cti_crossdataset_topk.json`, `results/cti_prospective_new_datasets.json`,

results/cti_prospective_k_sweep.json, results/cti_k_boundary_sweep.json, results/cti_global_vs_single_surgery.json, results/cti_beyond_1nn.json, results/cti_crossdomain_surgery.json, results/cti_nlp_linear_regime_surgery.json, results/cti_kappa_eff_identifiability.json, results/cti_2d_causal_surface.json, results/cti_extended_family_loao.json, results/cti_neuroscience_cadieu2014.json, results/cti_stringer2018b_validation.json, results/cti_allen_neuropixels_validation.json, results/cti_allen_all_sessions_complete.json, results/cti_spread_vs_K.json, results/cti_spread_vs_K_n10.json, results/cti_kappa_universal_comprehensive.json, results/cti_beta_constrained.json, results/cti_multi_k_sweep.json, results/cti_k10_kappa_sweep.json, results/cti_neff_pairwise_extended.json, results/cti_all_models_fit.json, results/dose_response_preregistration.json, results/cti_equicorrelation_deff.json, results/cti_equicorr_K_sweep.json, results/cti_allen_equicorrelation.json, results/cti_competitive_geometry_verification.json, results/cti_cross_family_equicorr.json, results/cti_competitor_weight_map.json, results/cti_alpha_kappa_rank_corr.json, results/cti_weight_map_transfer.json, results/cti_weight_map_transfer_prereg.json, results/cti_two_knob_identifiability_keff.json, results/cti_preregistered_causal_intervention.json, results/cti_cifar_antitriplet.json, results/cti_confusion_gumbel_test.json, results/cti_confusion_causal_prediction.json, results/cti_cross_arch_abcs.json, results/cti_phi_upgrade_pooled.json, results/cti_kappa_eff_held_out.json, results/cti_kappa_tournament.json, results/cti_j1j2_factorial_rct.json, results/cti_phi_jacobian_synthetic.json, results/cti_phi_jacobian_real.json, results/cti_abcs.json, results/cti_k_scaling_abcs.json, results/cti_bge_dbpedia14_abcs.json, results/cti_null_control_abcs.json, results/cti_utility_revised.json, results/cti_allen_equicorr_multiarea.json, results/cti_allen_multiarea_batch.json, results/cti_downstream_protocol_v3.json, results/cti_downstream_h3_n9.json, results/cti_encoder_loao.json, results/cti_scaling_dynamics.json. Pre-registration timestamps are verifiable via git log. Code is available at https://github.com/dl1683/ai-moonshots.

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A Derivation Details

Starting from the Gumbel race identity (Eq. 4), under nearest-class dominance:

$$q \approx \left[1 + (K-1)\exp\left(-\frac{\Delta_k}{s}\right)\right]^{-1}.$$

Taking logits:

$$\mathrm{logit}(q) = \frac{\Delta_k}{s} - \log(K-1).$$

With $\Delta_k = a_0 + a_1\kappa_{\text{nearest}} + o(1)$:

$$\mathrm{logit}(q) = \underbrace{\frac{a_1}{s}}_{\alpha}\kappa_{\text{nearest}} - \underbrace{1}_{\beta_\infty}\log(K-1) + \underbrace{\frac{a_0}{s}}_{C_0} + o(1).$$

Finite-data corrections renormalize $\beta_\infty = 1$ to $\beta \approx 0.746$, yielding Eq. 1.

B Pre-registration Deviation Log

Table 6: Pre-registration deviation log. Commits with timestamps are verifiable via `git log --oneline`.

Commit	Change	Justification
`8250bf1`	Dataset substitution: TREC (K=6) → dair-ai/emotion (K=6); PolyAI/banking77 → mteb/banking77	The original datasets require legacy loading scripts deprecated by the `datasets` library (`trust_remote_code=True` no longer accepted). Replacement datasets have identical K values; the pre-registered $A(K)$ predictions (1.951 and 1.569) are unchanged.
`6497907`	Stratified sampling in amazon_massive K=60 test; rare-class guard in `compute_q`	Random subsampling of 1500 from 1265 available stratified samples left some classes with < 2 members, causing `StratifiedShuffleSplit` to fail. The fix (draw $\geq$20 samples per class; drop classes with < 2 members before CV) is a numeric stability patch with no effect on the analysis model or the frozen $A(K)$ prediction.
`7f4c9f8`	RWKV acceptance interval $[2.43, 3.29]$: pre-registration JSON describes this as "mean $\pm$ 3*std(0.054)", but $3 \times 0.054 = 0.162$ gives $[2.727, 3.051]$, not $[2.43, 3.29]$.	The interval half-width 0.43 actually equals $\approx 8\sigma$ of the locked std, not 3σ. This is a documentation error in the pre-registration artifact; the locked interval $[2.43, 3.29]$ and the observed RWKV $\hat{\alpha} = 2.887$ are both as stated. The outcome is unaffected: 2.887 lies within any reasonable interval around 2.889 (the 11-arch prior mean). The paper text now correctly states "wider than $\pm 3\sigma$" (§5.2).

Neither edit changes the pre-registered predictions (A values, hypothesis thresholds, or evaluation metrics); both are verifiable as non-analysis-changing by inspecting the git diffs.

C Surgery Details

Initial attempts to increase κ_{nearest} via CE + triplet-loss fine-tuning with naïve $\lambda = 0.5$ caused representation collapse (within-class variance exploded, q dropped $0.70 \to 0.22$; the logged collapse metric was the Fisher scatter ratio S_B/S_W, which differs from κ_{nearest}). With L2-normalisation and $\lambda = 0.1$, a pre-registered causal intervention (commit `94ac8a3`; `cti_preregistered_causal_intervention.json`) produced a clean bidirectional training manipulation: CE arm $\text{mean}_q = 0.654$ vs. triplet arm $\text{mean}_q = 0.868$ ($\Delta q = +0.213$, 5/5 seeds direction-correct, PARTIAL PASS); an anti-triplet arm (commit `2b7a812`; `cti_cifar_antitriplet.json`) reduces q to 0.12–0.22 across all 5 seeds (below CE baseline). This bidirectional training manipulation independently confirms that κ_{nearest} is causally active in both directions. The quantitative CTI prediction underestimates by $\approx 5\times$ (predicted $\Delta\,\text{logit} = 0.225$, actual$= 1.244$): training manipulations alter the full representation geometry (not only κ_{nearest}), so the marginal $\alpha \cdot \Delta\kappa$ formula does not account for all confounds in a training-based setting. Single-direction frozen surgery (scaling ONE centroid-pair direction) underperforms the full CTI prediction by $\approx 16\times$: this is understood as a predicted $1/d_{\text{eff}}$ attenuation (a single dimension out of $d_{\text{eff}} \approx 24$). A pre-registered global surgery test (scaling ALL $K-1$ signal-subspace directions; commit `1abdef6`) recovers the full predicted effect in the *linear regime* (CIFAR-100, $\kappa_{\text{eff}} \approx 1.0$, 3 seeds): ratio $\Delta\,\text{logit}_{\text{global}} / \Delta\,\text{logit}_{\text{single}} = 18.74 \approx d_{\text{eff}} = 24.52$ (within-15% at $r = 1.5$). Two pre-registered cross-domain follow-ups on frozen NLP embeddings failed H1: (1) sub-linear regime ($\kappa_{\text{eff}} = 0.24$–$0.73$): median ratio$\approx 2 \ll d_{\text{eff}} \approx 33$; (2) pythia-1b, $\kappa_{\text{eff}} = 1.33$: median ratio$\approx 3.68 \ll d_{\text{eff}} = 54$. Global surgery consistently outperforms single-direction surgery in NLP (directional H2 passes: 10/13 and 5/5), but the quantitative d_{eff} prediction fails. The $1/d_{\text{eff}}$ attenuation mechanism appears specific to the trained CIFAR-100 CNN context.

D Two-Component Geometry Details

Null-space identifiability (commit `6938940`; `cti_kappa_eff_identifiability.json`): null-space scaling provably holds κ_{eff} invariant while changing κ_{nearest} ($< 0.001\%$ variation, H1: 3/3); q tracks κ_{nearest} (OLMo-1B: $r = 1.00$), not κ_{eff} (H4: 1/3 PASS).

2D causal surface (commit `2bc7da6`; `cti_2d_causal_surface.json`): a 3×3 factorial orthogonally manipulates both components on 2 NLP architectures. A bivariate model $\text{logit}(q) = \alpha_1\kappa_{\text{nearest}} + \alpha_2\kappa_{\text{eff}} + C$ does not significantly outperform the univariate κ_{nearest} law (H4 FAIL: $\Delta R^2 = 0.03/0.001$; $R^2 = 0.92$–0.98 for κ_{nearest} alone). Cross-architecture bivariate transfer is strong (H6: $r = 0.988, 0.967$). A dimensional confound limits the factorial: in $d = 2048$ with $\approx$2029 null dimensions, compensated signal-scaling redistributes high-dimensional null noise. Combined, these experiments establish κ_{nearest} as the dominant causal predictor; d_{eff} adds marginal information.

Two-knob identifiability (commit `7def904`; `cti_two_knob_identifiability_keff.json`): the most direct test of whether $\kappa_{\text{eff}} = \kappa_{\text{nearest}}\sqrt{d_{\text{eff}}}$ is the sufficient statistic. On CIFAR-100 ($K = 20$, linear-regime checkpoint, 3 seeds), two independent surgical knobs are applied: (A) centroid scaling to change κ_{nearest} while holding d_{eff} fixed; (B) within-class variance redistribution to change d_{eff} while holding κ_{nearest} fixed. If κ_{eff} were sufficient, matched pairs (same κ_{eff}, different κ/d_{eff} decomposition) should give equal $\text{logit}(q)$. **Result (6/7 criteria fail):** mean $|\Delta\,\text{logit}| = 0.30$ across 33 matched pairs (threshold ≤ 0.08); only 24% of pairs meet the invariance criterion (threshold $\geq 80\%$). Crucially, $\text{corr}(\text{residual}, d_{\text{eff}}) = -0.790$: the d_{eff} surgery systematically overpredicts q, while the κ_{nearest} surgery correctly tracks q. **Interpretation:** κ_{nearest} alone is the causal driver; the $\sqrt{d_{\text{eff}}}$ factor in the LOAO formula captures cross-architecture variation observationally but is not independently causally active when manipulated in isolation.

Table 7: Global fit statistics (single C_0 and per-dataset C_0).

Quantity	Single C_0	Per-dataset C_0
$\hat{\alpha}$	2.8660	1.4773
$\hat{\beta}$	0.7459	0.3262
R^2	0.6273	0.9552
LOAO CV_α	0.0191	0.0228
LODO CV_α	0.4193	–

Table 8: RWKV pre-registration and outcome.

Item	Value
Locked prior mean (11 arch)	2.8886
Locked prior std	0.0540
Pre-registered acceptance interval	$[2.43, 3.29]$
Observed RWKV LOAO α	2.8869
Outcome	**PASS**

E Additional Tables

F Competition Field: Extended Results

F.1 Competition field: j2 is causally active and effects are additive

The original CTI law uses only κ_{j1} (nearest competitor). A 6-arm pre-registered RCT on frozen DBpedia-14 embeddings (pythia-160m, $K = 14$) revealed that the *second-nearest* competitor j_2 is also causally active: single-competitor shift with j_1 fixed gives $r(\kappa_{j2}, \operatorname{logit} q) = 0.811$, exceeding the $r < 0.20$ sufficiency threshold. The full competition field $r(\kappa_{\text{all}}, \operatorname{logit} q) = 0.988$.

Upgraded law via soft-min competition field. We fit $\phi(\tau) = -\tau \log \sum_{j \neq i} \exp(-\kappa_{ij}/\tau)$ (log-sum-exp soft-min of all competitors) as a replacement predictor for κ_{j1}. A pooled tau-sweep across 5 architectures (70 data points, `cti_phi_upgrade_pooled.json`) finds $\tau^* = 0.20$ (best R^2), yielding $R^2 = 0.461$ vs. $R^2 = 0.360$ for κ_{j1} alone (+28.1% relative improvement), with an *out-of-sample* test on 4 held-out architectures confirming the transfer: pooled R^2 improves from 0.451 to 0.578 (+28.3%, 4/4 architectures pass, `cti_kappa_eff_held_out.json`).

Limitation (functional form specificity): A pre-registered tournament test compared causal weights $\{w_r\}$ against 1000 random monotone weight assignments. The causal weighting achieves $R^2 = 0.578$ and exceeds 67.6% of random nulls, but does not exceed the 90th-percentile random baseline ($R^2_{p90} = 0.585$; pre-registered threshold; FAIL, `cti_kappa_tournament.json`). The *direction* of the competition gradient (near competitors matter more) is validated; the *specific* exponential functional form $\exp(-\text{gap}/\tau^*)$ is not yet decisively distinguished from arbitrary monotone weighting on this dataset.

Additivity RCT. A pre-registered factorial RCT (`cti_j1j2_factorial_rct.json`) jointly perturbed j_1 and j_2 over DELTA $\in \{0.0, 0.1, 0.2, 0.3, 0.5, 0.8, 1.0\}$. *Additivity*: $r(\Delta \operatorname{logit}_q, \Delta\kappa_{j1} + \Delta\kappa_{j2}) = 0.9964$ (n=84 class-delta pairs), confirming effects sum linearly. Mean relative weight $\bar{w} = 0.569$ (Codex prediction: 0.40; threshold: $|w - 0.40| < 0.20$; PASS). Near-tied pairs (gap≈0.004) yield $w \approx 6.4$, while wide-gap pairs give $w \approx 0$, consistent with the phi model: $w \propto \exp(-\text{gap}/\tau^*)$.

Synthetic Jacobian validation. A synthetic Gaussian experiment ($K = 14$, $d = 50$, $N = 2000$; `cti_phi_jacobian_synthetic.json`) fixes $\kappa_{j1} = 0.40$ and varies gap $\in$

$\{0.15, 0.25, 0.35, 0.50, 0.70, 1.00\}$. Results: $r(\log w, -\text{gap}) = 0.9502$ (threshold > 0.85; PASS); fitted $\tau^*_{\text{median}} = 0.207$ (empirical $\tau^* = 0.20$; error 3.5%). This provides a first-principles derivation of τ^* from the geometry.

Real-embedding Jacobian (inconclusive): A matched-delta Jacobian test on pythia-160m step-512 checkpoint embeddings (`cti_phi_jacobian_real.json`) found only 2 of 14 classes meeting both non-ceiling ($q < 0.85$) and rank-stable (gap> 0.04) criteria; the test was inconclusive due to insufficient regime coverage. Qualitatively, wide-gap classes (gap $>$ 0.50) correctly show $w \approx 0$, consistent with phi model predictions; quantitative validation of the exponential functional form in real embeddings remains open.

Confusion-matrix Gumbel test. We test the Gumbel race mechanism directly via the 1-NN confusion matrix C_{ij} (fraction of class i examples whose nearest neighbor is from class j, estimated by 5-fold cross-validation). The Gumbel race predicts $\log C_{ij} \approx \text{const}_i - \kappa_{ij}/\tau_{\text{err}}$, i.e. confusion decays exponentially with kappa (`cti_confusion_gumbel_test.json`). **Pre-registered H1 PASS**: pooled Pearson $r(\log C_{ij}, -\kappa_{ij}) = 0.750$ ($n = 151$ non-zero pairs, threshold 0.50). **Pre-registered H4 PASS**: pooled Spearman $\rho(-\kappa_{ij}, C_{ij}) = 0.768$ (threshold 0.50). The test reveals a *two-level tau structure*: (1) within each class, confusion drops sharply (per-class median $\tau_{\text{err}} \approx 0.026$, winner-take-all: only the nearest competitor causes appreciable confusion); (2) across classes, the pooled best-fit temperature is $\tau_{\text{pooled}} = 0.201 \approx \tau^*$, consistent with the phi model's $\tau^* = 0.20$. This resolves an apparent tension: $\tau^* = 0.20$ captures between-class scaling (why low-kappa classes are confused more) while $\tau_{\text{err}} \approx 0.025$ captures the within-class sharpness (why κ_{nearest} alone is the sufficient statistic). H2/H3 (requiring per-class $\tau_{\text{err}} \in [0.10, 0.50]$) fail, confirming the local competition is sharper than the phi model's global fit suggests.

Causal confusion-matrix prediction (pre-registered). We run the decisive causal test proposed by Codex: shift the nearest competitor j_1 by $\delta \in \{1.0, 2.0, 3.0\}$ embedding units, predict the *full* confusion-matrix update using *fixed* $A = 1.054$ and $\tau^* = 0.20$ (no refit), then compare to the measured post-intervention confusion (`cti_confusion_causal_prediction.json`).

Pre-registered H1: pooled $r(\hat{C}_{\text{pred}}, C_{\text{actual}}) > 0.50$ using $\tau^* = 0.20$. Pre-registered H3: sign accuracy of $\Delta C_{i,j1}$ (predicted direction of confusion change) ≥ 0.80.

All criteria pass at all three deltas:

- $\delta = 1.0$: $r_{\tau^*} = 0.842$ (PASS), sign acc.$= 92.9\%$ (PASS)
- $\delta = 2.0$: $r_{\tau^*} = 0.808$ (PASS), sign acc.$= 100\%$ (PASS)
- $\delta = 3.0$: $r_{\tau^*} = 0.776$ (PASS), sign acc.$= 100\%$ (PASS)

Importantly, $\tau^* = 0.20$ outperforms $\tau_{\text{local}} = 0.026$ for causal prediction ($r = 0.842$ vs $r = 0.683$ at $\delta = 1.0$), confirming that τ^* captures causal response sensitivity while τ_{local} describes the equilibrium confusion structure. Predicted per-class q values match observed within 0.01–0.07 logit units (no refit), providing quantitative causal-predictive validation of the CTI framework.

Unconfounded competitor weight map (ci-fixed interventions). The competition experiments above shift c_j while c_i may move under projection; a fully unconfounded estimate requires anchoring c_i at its original position while moving only c_j. We run 14 classes $\times$ 13 competitors $\times$ 11 delta points $\times$ 5-fold CV with c_i fixed (`src/cti_competitor_weight_map.py`; `results/cti_competitor_weight_map.json`). The per-rank causal weight $w_r = \alpha_r/\alpha_{j_1}$ (normalised by nearest-competitor slope) yields the profile $w = [1.00, 0.29, 0.21, 0.17, 0.09, 0.08, 0.03, \ldots]$: a heavy-head distribution where only ranks 1–4 are causally significant. The summed effective competition $\sum_r w_r = 2.42$ is consistent with $\alpha_2 = 1.523$ (2-competitor localisation; §F.1), independently confirming that 2–4 near competitors drive the cross-architecture constant. The nearest-competitor slope $\alpha_{j_1} = 1.272$ is close to (but below) LOAO $\alpha = 1.477$; the gap reflects residual saturation in high-accuracy DBpedia classes.

Alpha-kappa rank correlation: monotone decay confirmed. As an internal validation that competition weight decays with kappa distance (as theory requires), we compute Spearman $\rho(\alpha_r, \text{kappa_rank})$ for each of the 14 classes (`results/cti_alpha_kappa_rank_corr.json`). **14/14 classes show negative correlation** (sign test $p = 6.1 \times 10^{-5}$, Wilcoxon $p = 1.2 \times 10^{-4}$); mean $\rho = -0.551$, 95% CI $[-0.665, -0.436]$. This confirms that competition weight decays monotonically with kappa distance, as predicted by the Gumbel race theory, without relying on any functional-form assumption.

Cross-architecture weight transfer (pre-registered; negative result). We pre-registered a no-refit transfer test: use the pythia-160m weight profile w_r to predict total competition effects $\alpha_{\text{top}}\, m$ on two held-out architectures (pythia-410m, bert-base; `results/cti_weight_map_transfer_prereg.json`, pre-registered commit `c013ddc`). Both H1 (relative error $< 30\%$) and H2 (Pearson $r > 0.90$) fail on both architectures. Root cause: the held-out architectures operate in a saturation regime ($q > 0.85$) on DBpedia K=14, making $\alpha_{j_1} = \delta\,\text{logit}/\delta\kappa \to \infty$ and rendering the normalised weight profile unidentifiable. The pythia-160m source slope $\alpha_{j_1} = 1.272$ vs. pythia-410m $\alpha_{j_1} = 12.6$ and bert-base $\alpha_{j_1} = 9.6$ reflects saturation amplification rather than a genuinely different competition structure. We record this as a pre-registered negative result: w_r cross-architecture universality cannot be evaluated on a dataset where held-out models are near ceiling. The finding motivates future transfer tests on harder datasets ($q \in [0.5, 0.8]$).

Table 9: Competition field evidence summary.

Experiment	Result	Status
j2 causal activity (6-arm RCT)	$r = 0.811$	Causally active
Full field activity	$r = 0.988$	PASS
Phi upgrade pooled R^2	$0.360 \to 0.461$ (+28.1%)	PASS
Held-out arch transfer (4/4)	$0.451 \to 0.578$ (+28.3%)	PASS
Monotone null (causal vs random weights)	67.6% beat, p90 FAIL	FAIL
Additivity RCT	$r = 0.9964$	PASS
Synthetic Jacobian $r(\log w, -\text{gap})$	0.9502	PASS
Synthetic τ^*	0.207 vs. empirical 0.20	3.5% error
Real-embedding Jacobian	2/14 valid classes	INCONCLUSIVE
Confusion matrix Gumbel (r, ρ)	0.750, 0.768	PASS (H1+H4)
Causal confusion prediction ($\delta = 1..3$, no refit)	$r = 0.842$–0.776	PASS (H1+H3)
Unconfounded weight map (ci-fixed)	$\sum_r w_r = 2.42$; w_1–w_4 significant	PASS
α-kappa rank correlation	14/14 negative, $\bar{\rho} = -0.551$, $p = 1.2 \times 10^{-4}$	PASS
Cross-arch weight transfer (pre-reg.)	H1/H2 FAIL (saturation regime)	NEGATIVE
Cross-arch A1 scaling (5 archs, 3 families)	r-scale ≥ 0.90 (5/5; min observed 0.961)	PASS
Null-space surgery control	slope ratio 1.4% of competitive	PASS
K-scaling ABCS ($K = 6, 9, 10, 20$)	$r(K, K_{\text{eff}}) = 0.998$; A1 4/4	PASS
BGE-small encoder ABCS ($K = 14$)	r-scale $= 0.981$; $K_{\text{eff}} = 13$	PASS

G Composite Kappa-K Sweeps: Detailed Results

G.1 Composite Kappa-K Law: Multi-K and K=10 Sweeps

The comprehensive universality test (above) sweeps architecture at fixed K per dataset. A complementary validation sweeps κ at fixed K: how does q change as the nearest-class gap varies, and does the $\sqrt{K-1}$ competition scaling persist at different K?

B(κ) power-law form. Define $B = q_{\text{base}}/(1 - q_{\text{base}})$ as the baseline performance ratio when only the minimum κ class pair is varied. The Gumbel race predicts:

$$B(K, \kappa) = C_0 \, \frac{\sqrt{K-1}}{\kappa^p}, \quad p > 0, \quad C_0 = \sqrt{2/3} \approx 0.816, \tag{7}$$

where p is the empirical kappa exponent and C_0 is derived analytically from the Gumbel race variance.

Table 10: Beta-constrained model comparison: LOAO and LODO MAE across 444 data points, 19 architectures, 10 datasets. Winner (M2, $\beta = 0.5$) is highlighted.

Model	β	LOAO MAE	LODO MAE	Status
M2 (sparse)	**0.5**	**0.766**	**0.857**	**WINNER**
M0 (unconstrained)	0.478	0.771	0.898	+0.6% vs M2
M3 (K-indep.)	0.0	0.917	0.899	
M1 (dense ETF)	1.0	0.852	1.069	+10.1% vs M2

Multi-K sweep ($K \in \{5, 7, 10, 14, 20\}$, pre-registered). The pre-registered multi-K sweep (`src/cti_multi_k_sweep.py`) varies K at a fixed target κ and measures B. Result (`results/cti_multi_k_sweep.json`): $B \propto \sqrt{K-1}$ with CV $= 4.4\%$ across 5 K values and 3 alpha levels — the $\sqrt{K-1}$ scaling is confirmed with high consistency. This is independent evidence for $N_{\text{eff}} \propto \sqrt{K-1}$ using synthetic Gaussian-mixture embeddings (not real neural networks).

K=10 kappa sweep (pre-registered). The $K = 10$ sweep (`src/cti_k10_kappa_sweep.py`, 6 kappa values $\kappa \in \{0.15, 0.20, 0.25, 0.30, 0.40, 0.50\}$, 3 alpha levels, 12 seeds) tests whether Eq. 7 holds at $K = 10$ with the specific C_0 derived from the $K = 7$ strong pass.

Results (`results/cti_k10_kappa_sweep.json`): **STRONG PASS** (all 5 pre-registered criteria):

- **PR1** (all $|\text{rel_err}| < 30\%$): 6/6 kappas pass. Maximum relative error $= 19\%$ (kappa$=$ 0.20).
- **PR2** (log-log slope $\in [-0.90, -0.55]$): slope$= -0.739$. PASS.
- **PR3** (C_0 within $\pm 20\%$ of prior 0.816): $C_0 = 0.773$, relative error $= 5.4\%$. PASS.
- **PR4** (C_0 CV $< 25\%$): CV$= 4.2\%$. PASS.
- **PR5** (Pearson $p < 0.05$): $r = -0.992$, $p < 10^{-4}$. PASS.

The $C_0 = 0.773$ at $K = 10$ is 5.4% below the prior $C_0 = 0.816$ (K=7 result), consistent with easy-class dilution: at $K = 10$, 8 easy classes with large κ drive q_{base} near 1, diluting the κ-surgery signal. This systematic dilution decreases with K (more hard pairs at large K) and with κ (harder pairs have larger signal). The pattern is theoretically expected and confirms the law's functional form without altering the universal scaling.

H Sparse Competition: Full Analysis

H.1 Sparse Competition: Pre-Registered Beta Exponent Test

The Gumbel race identity (Eq. 4) under nearest-class dominance predicts $\beta_\infty = 1.0$ (all $K-1$ competitors equally active). The empirical global estimate $\hat{\beta} = 0.478$ (positive sign convention; from the 444-point fit) is substantially below 1.0. We tested two competing mechanistic hypotheses:

- **Dense ETF competition (M1)**: $\beta = 1.0$ (all $K-1$ competitors contribute equally, as in an equiangular tight frame); α and C_0 fit freely.
- **Sparse competition (M2)**: $\beta = 0.5$ (only $N_{\text{eff}} \propto \sqrt{K-1}$ competitors contribute); α and C_0 fit freely.

We additionally tested M0 (all free) and M3 ($\beta = 0$, K-independent).

Pre-registration. The model comparison was pre-registered (`src/cti_beta_constrained_test.py`, committed before running; `results/cti_beta_constrained.json`) with criteria: (PR_WIN) M2 LOAO MAE $\leq$ M1 LOAO MAE; (PR_CLOSE) relative improvement $\geq 3\%$; (PR_UNCON_GAP) unconstrained M0 within 5% of winner.

All three pre-registered criteria pass. PR_WIN: M2 MAE $= 0.766 <$ M1 MAE $= 0.852$ (10.1% improvement). PR_CLOSE: PASS ($\geq 3\%$ threshold). PR_UNCON_GAP: M0 (unconstrained, $\hat{\beta} = 0.478$) is within 0.6% of M2, confirming $\beta = 0.5$ is not under-constrained—it is effectively the best global estimate. LODO confirms: M2 beats M1 by 24.8% (0.857 vs. 1.069).

Per-K winner breakdown. M2 ($\beta = 0.5$) wins at $K \in \{4, 6, 10, 59, 77\}$ (sparse, task-specific classes); M1 ($\beta = 1.0$) wins at $K \in \{14, 20, 28, 42\}$ (semantically dense topic/emotion datasets). This is consistent with a regime-dependent competition structure: at low K (few distinct classes) and very high K (many fine-grained task classes), competition is sparse ($\sqrt{K}$ scale); at medium K with semantically overlapping topics (news, emotions), competition approaches the dense ETF limit.

Sparse competition mechanism. The Gumbel race sum (Eq. 4) can be decomposed as $\sum_{j \neq k} \exp(-\Delta_{k,j}/s) \approx N_{\text{eff}} \exp(-\Delta_k/s)$ where N_{eff} is the effective number of competitive classes. The empirical $\beta = 0.5$ implies $N_{\text{eff}} \propto \sqrt{K-1}$. A plausible mechanism (Codex analysis): in real text classification, classes are organized hierarchically. Competitors at the same semantic level number $\sim \sqrt{K}$; deeper-level competitors are substantially farther away and contribute negligibly. Equivalently, if each competitor activates an effective gate with probability $\propto 1/\sqrt{K}$ (sparse gating), then $N_{\text{eff}} = (K-1) \cdot p_K \approx \sqrt{K-1}$.

Triangulated evidence for $N_{\text{eff}} \propto \sqrt{K-1}$. Three independent experimental frameworks converge on the same exponent:

1. **Synthetic sweep (controlled geometry)**: the multi-K sweep (Eq. 7, 5 values $K \in \{5, 7, 10, 14, 20\}$) finds $B(K, \kappa) \propto \sqrt{K-1}/\kappa^p$ with CV= 4.4%. The numerator $\sqrt{K-1}$ arises directly from the Gumbel race competition count.
2. **Real network global fit**: across 444 points (19 architectures, 10 datasets, $K = 4$–77), the unconstrained OLS estimate is $\hat{\beta} = 0.478 \approx 0.5$.
3. **Pre-registered constrained comparison**: $\beta = 0.5$ beats $\beta = 1.0$ by 10.1% LOAO and 24.8% LODO; the unconstrained global estimate lies within 0.6% of $\beta = 0.5$, confirming it is the MLE under the Gumbel race model.

Synthetic and real-network results are mutually independent: (1) uses Gaussian-mixture embeddings with controlled κ/d_{eff}, while (2)–(3) use transformer embeddings on natural-language tasks.

Table 11: N_{eff} geometric exponent and competition density $\rho = N_{\text{eff}}(K{=}14)/(K-1)$ across 7 architectures on DBpedia. Denser inter-class competition (higher ρ) predicts steeper N_{eff} scaling (higher β_{geom}). RWKV (pure linear RNN) validates the mechanism beyond attention-based models.

Architecture	Type	β_{geom}	$\rho = N_{\text{eff}}/(K-1)$
Pythia-160m	Decoder (GPT-NeoX)	0.934	0.804
TinyLlama-1.1B	Decoder (LLaMA)	0.923	0.779
RWKV-169M	Linear RNN	0.861	0.659
OLMo-1B	Decoder (OLMo)	0.806	0.578
Qwen2.5-0.5B	Decoder (Qwen)	0.829	0.564
Qwen3-0.6B	Decoder (Qwen3)	0.791	0.566
GPT-Neo-125M	Decoder (GPT-Neo)	0.711	0.454
Pearson $r(\rho, \beta_{\text{geom}}) = 0.978$ ($p = 0.0001$, $n = 7$); Spearman $\rho = 0.893$ ($p = 0.007$)			

Direct centroid-geometry and beta reconciliation. A direct test (`src/cti_neff_pairwise.py`) extracts all class centroids from Pythia-160m on DBpedia ($K = 4$–14 subsets, $n = 20$ seeds each) and computes $N_{\text{eff,k}} = \sum_{j \neq k} \exp(-\alpha_{\text{univ}}(\kappa_{kj} - \kappa_{k,\text{near}}))$ where $\alpha_{\text{univ}} = 3.598$. Results: DBpedia geometric exponent $= 0.934 \pm 0.005$ ($r = 0.9999$); Banking77 ($K = 4$–64) exponent $= 0.918 \pm 0.012$ ($r = 0.9999$). Across seven architectures

(Pythia-160m, TinyLlama-1.1B, RWKV-169M, OLMo-1B, Qwen2.5-0.5B, Qwen3-0.6B, GPT-Neo-125M), the geometric N_{eff} exponent ranges from 0.71 to 0.93 (mean 0.84; see Table 11 and `results/cti_neff_pairwise_extended.json`). The exponent correlates with competition density $\rho = N_{\text{eff}}(K_{\text{full}})/(K_{\text{full}} - 1)$ at the full class count: Pearson $r = 0.978$ ($p = 0.0001$, $n = 7$), Spearman $\rho = 0.893$ ($p = 0.007$). Architectures that create denser, more uniform inter-class separation (Pythia-160m: $\rho = 0.80$, TinyLlama-1.1B: $\rho = 0.78$) exhibit steeper geometric scaling; those with sparser competition (GPT-Neo-125M: $\rho = 0.45$) scale more sub-linearly. Crucially, RWKV (pure linear RNN with no attention) follows the same trend ($\rho = 0.66$, $\beta_{\text{geom}} = 0.86$), confirming the mechanism is architecture-independent.

These geometric exponents (0.71–0.93) exceed the regression-based $\beta = 0.478$; this does *not* represent a contradiction. A direct within-architecture test (`src/cti_beta_reconciliation.py`) measures the univariate slope of $\text{logit}(q_{\text{norm}})$ vs $\log(K - 1)$ as K is varied by subsampling classes from a fixed architecture (Pythia-160m). Without controlling for κ_{nearest}: slope ≈ -0.52 (DBpedia), -0.41 (Banking77) — both consistent with $\beta \approx 0.5$. After controlling for κ_{nearest}: slope $\approx +0.03$ (DBpedia), $+0.31$ (Banking77) — κ_{nearest} mediates nearly all K-dependence within a single architecture, especially at small K ranges. The cross-dataset regression $\beta = 0.478$ is an average partial slope across datasets with varied C_0; the geometric N_{eff} exponent measures a distinct quantity (competition density scaling) that is inherently above β because it does not subtract the κ_{nearest} mediation term. Both quantities confirm the sparse-competition regime ($\beta < 1$): the Gumbel race with $N_{\text{eff}} \propto K^{0.81}$, combined with κ_{nearest} mediating K-dependence, produces the observed $\beta \approx 0.5$ in the law.

H.2 H8+ expanded holdout detail

Table 12: H8+ per-model holdout summary (11 unseen architectures). All parameters frozen from 20-model training set (pre-registered commit `719da25`). n is the number of valid dataset predictions per model after floor/ceiling exclusion.

Model	Family	n	MAE	Max err	Status
distilbert-base-uncased	encoder	8	0.036	0.060	Best
albert-base-v2	encoder	7	0.067	0.122	Good
roberta-base	encoder	8	0.071	0.309	Good
falcon-rw-1b	decoder	8	0.051	0.110	Good
stablelm-3b-4e1t	decoder	8	0.070	0.155	Good
opt-125m	decoder	7	0.071	0.148	Good
qwen2.5-1.5b	decoder	7	0.077	0.194	Good
phi-1.5	decoder	8	0.082	0.183	Good
pythia-2.8b	decoder	8	0.082	0.172	Good
bloom-560m	decoder	4	0.160	0.316	Weak
gemma-3-1b	decoder	4	0.167	0.533	Weak
All (full H8+)		77	0.077	0.533	PASS
Excl. bloom+gemma		69	0.067	0.309	—
$q \in [0.05, 0.95]$		73	0.073	0.533	—

bloom-560m operates near chance on 4/8 datasets (all excluded by floor filter or yielding large errors); gemma-3-1b catastrophically mispredicts DBpedia (err $= 0.533$, predicted $q = 0.681$, actual $q = 0.148$). These two models account for all outliers > 0.20. Excluding them yields $r = 0.938$, MAE$= 0.067$ ($n = 69$). The three newly added models (falcon-rw-1b, phi-1.5, qwen2.5-1.5b) all perform well (MAE ≤ 0.082), confirming that the law generalizes beyond the initial 8-model set. The pre-registered test reports the full set; the sensitivity rows are exploratory.